

A Theory of Optimal Disaster Insurance under Intermediary Constraints

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Abstract

Implicit insurance subsidies in housing markets protect households from rising weather damage but constrain insurers financially, undermining the long-run availability of insurance. Using a dynamic housing model calibrated to the California housing market, I characterize the optimal time-consistent insurance allocation that protects housing values while limiting insurance market crashes after disaster shocks. The severity of these crashes is governed by the economy-wide marginal propensity to insure. I decentralize the social planner's optimal allocation with an insurance tax whose sign varies with household leverage and the likelihood and cost of binding insurer capital constraints in the future. I further provide empirical evidence consistent with insurer financing constraints amplifying insurer withdrawals from insurance markets.

JEL Codes: G22, Q54, R31, G28, D62

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1 Introduction

Property insurance markets are under stress due to growing economic damages from natural disasters that create record-level claim payouts. Following disaster losses, the availability of insurance has declined as private insurers started to exit disaster-prone areas. Such insurance contractions make homeowners more exposed to natural disasters, affect house prices ([Eastman et al., 2024](#)) and potentially impose significant macroeconomic costs ([ECB-EIOPA, 2023](#)).

To keep insurance available and affordable, U.S. regulators have intervened across several dimensions, including by capping rate increases, providing insurance through residual insurers to homeowners who cannot obtain coverage in the private market, using state guaranty funds to pay claims when insurers default, and directly subsidizing insurance. While mitigating under-insurance, these policies could crowd out the long-run availability of private insurance by undermining insurers’ financial conditions.¹

A fundamental question is therefore which regulation optimally balances the short-term need to keep homeowners insured with insurers’ financial stability, a condition for keeping homeowners insured in the long run. The literature has made progress in estimating the sign and magnitude of regulatory distortions in insurance markets, yet lacks a structural analysis of (i) the conditions under which regulatory interventions are justified and (ii) how optimal regulation depends on the joint state of the insurance and housing markets.

In this paper, I develop a dynamic housing model with insurance intermediaries, calibrated to the California housing market, to characterize the optimal time-consistent insurance policy that jointly mitigates under-insurance and insurance market crashes. Two occasionally binding constraints are central to the model’s inefficiencies that motivate regulatory intervention: a household collateral constraint and an insurer capital constraint. Disaster shocks destroy collateral and can lead to binding collateral constraints when insurance take-up is low. Upon binding collateral constraints, house prices drop sharply as households need to deleverage, destroying borrowing capacity and ultimately consumption. Similarly, disaster shocks lead to sharp reductions in the availability of insurance when the insurer’s capital constraint binds, as insurers need to deleverage.

In a decentralized equilibrium, households take house prices and insurer net worth as given. Consequently, they do not internalize how insurance decisions today affect next-period

¹[Oh et al. \(2021\)](#) and [Liu and Liu \(2024\)](#) show that insurance rate caps are associated with an increase in insurer exits. [Abramson et al. \(2025\)](#) show that underpriced government policies can crowd out private insurers. [Sastry et al. \(2025b\)](#) show that the presence of state guaranty funds increases incentives for insurers to exit the market.

house prices and borrowing capacity upon disaster shocks when the collateral constraint becomes binding. Similarly, households do not internalize how high insurance demand today affects future insurance availability after disaster shocks when capital constraints become binding.

A social planner who faces the same constraints as the household internalizes these externalities and optimally manages both of the occasionally binding constraints. I decentralize the optimal policy with a subsidy on insurance, which can be either positive or negative, which, in the latter case, implies taxing insurance. The tax varies with two distinct characteristics of the homeowners insurance market. First, the optimal tax depends on how house prices change with insurance coverage, both directly and indirectly through insurance protecting collateral values in the event of disasters. Because insurance prevents house price declines upon a disaster and preserves consumption, a subsidy is justified.

Second, the optimal tax depends on how much private markets contract upon a disaster shock. Overinsurance in good times could lead to steep contractions in bad times when weather risks materialize. To quantify this, I introduce the concept of insurers' marginal propensity to insure (MPI), which characterizes insurers' risk-bearing capacity. MPI captures the sensitivity of the quantity of insurance policies sold to changes in insurer net worth. The concept thus reflects how well incumbent insurers are able to absorb disaster shocks and continue to supply insurance. MPI depends on capital constraints and disaster risk, but also on insurer investment returns, which act as a substitute for the insurer's product portfolio. When MPI is low, future insurance supply responds little to net worth shocks. The economy-wide MPI is an important concept in intertemporal trade-offs in insurance markets: insuring disaster risks provides macroeconomic benefits in the short-run, but if the MPI curve is sufficiently steep, high insurance take-up today could jeopardize the long-run availability of insurance through the financial constraints of the insurer, justifying a tax on insurance.

In the calibrated model, I find that the optimal macro-prudential insurance subsidy averages 4.8%, though in the simulation it varies over time with the joint state of the housing and insurance markets. The optimal insurance subsidy is close to zero when household indebtedness is low. Low debt states rarely feature housing fire sales upon disasters, reducing the need to incentivize insurance take-up. Conversely, insurance subsidies increase in household indebtedness. Insurance subsidies are also lower when the insurance market is fragile. When insurers have low net worth, there is a high likelihood that insurers face binding capital constraints when a disaster materializes, in which case insurance supply contracts and

insurance prices increasingly reflect financial frictions of the insurer. This state of the insurance market is costly for households and a lower insurance subsidy reduces the likelihood of reaching these states.

Empirically, I provide new motivating evidence on how insurance contractions transmit into housing markets. First, I document that insurer financial constraints have contributed to insurance contractions. Contractions in the zip codes with highest fire risk in California were especially steep when the zip code was served by insurers with low capital ratios. Second, instrumenting insurance prices and non-renewals with out-of-zip-code disaster damages, I find that, conditional on weather risk, insurance price increases and non-renewals negatively affect house prices. Third, disaster shocks lower house prices in the struck regions especially when insurance coverage is low and household indebtedness is high, a dynamic that the theoretical model replicates.

This paper makes three contributions. First, I contribute to the literature studying the effects of insurance market regulation. Capital regulation can distort insurance pricing and supply (Kojen and Yogo, 2015), affect insurers’ appetite for insuring natural disaster risk (Born and Klimaszewski-Blettner, 2013), and affect insurers’ risk-taking (Becker and Opp, 2013). Insurance rate caps can drive insurer exits (Oh et al., 2021, Liu and Liu, 2024). While the literature estimates regulatory distortions in reduced form, structural work is needed to understand which conditions justify insurance regulatory intervention and which regulation is optimal in disaster insurance markets. My main contribution is deriving the optimal time-consistent insurance policy in a dynamic housing model calibrated to the California housing market and providing state-dependent insurance regulation that jointly manages household collateral and insurer capital constraints.

Second, my paper contributes to the literature that investigates demand- and supply-side frictions as drivers of equilibrium insurance quantities. On the demand side, financial constraints (Sastry et al., 2025a) and behavioral explanations such as tail-risk neglect (Kunreuther and Pauly, 2004), learning (Gallagher, 2014) and limited attention (Cookson et al., 2024) explain insurance take-up. On the supply side, institutional constraints (Kojen and Yogo, 2022), regulatory frictions (Oh et al., 2021), informational frictions (Boomhower et al., 2024), capital market frictions (Eastman et al., 2024, Solomon, 2024) and coordination failures between insurers (Ibragimov et al., 2009) affect insurance supply. Evaluating the equilibrium effects of regulatory interventions comprehensively requires modeling how they affect insurance demand and supply simultaneously. To this end, I develop a dynamic model with occasionally binding constraints on both the household and intermediary side.

Third, I contribute to the literature studying how financial intermediaries shape macroeconomic fluctuations. The key insight in this literature is that financial frictions among intermediaries can amplify aggregate fluctuations. So far, the literature connecting intermediaries to business cycle fluctuations has focused on banks (Gertler and Kiyotaki, 2010, Coimbra and Rey, 2024, Favara and Imbs, 2015, Jamilov and Monacelli, 2025). Because insurance markets have first-order effects on housing markets, which are closely connected to real economic activity, it is important to study how insurers affect the broader economy.² Property insurers furthermore fulfill a critical role in society by allowing households and firms to offload risks, potentially affecting consumption and investment through precautionary motives, and enabling financial contracts, such as mortgages and firm loans. This paper develops a dynamic stochastic general equilibrium model in which insurance markets, housing markets, and aggregate consumption co-move, which allows for understanding how insurance market conditions affect the real economy.

2 Motivating Empirical Evidence

I start by providing empirical facts on contractions in weather-prone insurance markets. I then provide new empirical analyses that quantify the relevance of insurance market frictions for macro-economic outcomes. I specifically show that contractions in homeowners insurance coverage does not solely imply a transfer of weather risk from insurer to households, but can have first-order consequences in housing markets. Quantifying these effects are a first important step in quantifying welfare benefits of public insurance interventions that aim to curb under-insurance. Lastly, I provide insight into the role of supply-side frictions in insurance market contractions.

Fact 1: The steepness of insurance market contractions increases with insurer financial constraints.

I begin by showing trends in two insurance markets highly exposed to climate risk: California and Florida. Each year, I sort Floridan counties and Californian zip-codes into quintiles based on their hurricane and fire risk score, respectively.³ Subsequently, within

²Property insurers in the U.S. provide over \$15 trillion in multi-peril insurance coverage annually to homeowners.

³County-level hurricane risk scores are time-invariant, while zip-code fire risk scores vary over time. To standardize the sorting procedure, I average fire risk scores by zip-code. Averaging risk scores ensures that the composition of sorts does not change, preventing sudden changes in trends that result from changing portfolio compositions rather than market conditions.

each hurricane or fire risk quintile, I sort regions into two groups based on the incumbent insurers’ average risk-based capital ratios (RBC), weighted by their market shares. The county and zip-code average insurers’ RBC ratios are based on the entire sample period and the sorts are thus time-invariant. The double sort allows me to assess the relevance of insurer financial conditions in the steepness of insurance contractions after conditioning on risk exposure. Figure 1 shows the time trend in policies issued by private insurers, the market share of the residual insurer and average insurance premiums in both California and Florida for the each double sort.

The left panels reveal a sharp decline in privately issued insurance policies relative to the benchmark year 2009. While private market participation in low-risk segments in California has remained relatively stable or increased, the number of privately insured homeowners reduced sharply in both low- and high-risk segments. Contractions in the regions with highest weather risk were especially steep when the region was occupied by insurers with low capital ratios, which suggests that financial constraints playing a role in the steepness of insurance contractions. Even in low risk regions, growth was less or contractions were steeper when the region was served primarily by insurers with low capital ratios. Strikingly, even Floridan counties with low risk scores saw major exits, suggesting relatively low risk counties are also deemed to risky to insure.

The center panels indicate that conditional on risk exposure, the market share of the residual insurer is high when the region is occupied by insurers with low RBC ratio. This pattern is consistent across both regions and suggests that insurers financial constraints have contributed to the growth of residual markets. For example, in 2022, the share of the residual insurer FAIR plan in California in zip-codes with the highest fire risk was 40% when the zip-code was occupied by insurers with high capital ratios and 60% when occupied by insurers with low capital ratios.

The right panels show that homeowners insurance premiums have been rising sharply, primarily in high-risk segments. In Florida, average insurance premiums increase in insurer financial constraints. In California, in contrast, conditional on the region’s weather risk, average insurance premiums decrease with insurer financial constraints. If insurer financial constraints correlate negatively with informational advantages, this is consistent with less-informed insurers underpricing informed insurers as in [Boomhower et al. \(2024\)](#).

Table A.3 formally tests the role of insurer financial conditions in increasing insurance premia and non-renewals in California. Columns 1-4 regress the change in zip-code level average insurance premia on changes on average insurer net worth and RBC ratios. Columns

Homeowners Insurance Market Contractions by Risk and Insurer Financial Conditions

·· Low RBC (Q1) — High RBC (Q5) — Low Risk (Q1) - High Risk (Q5)

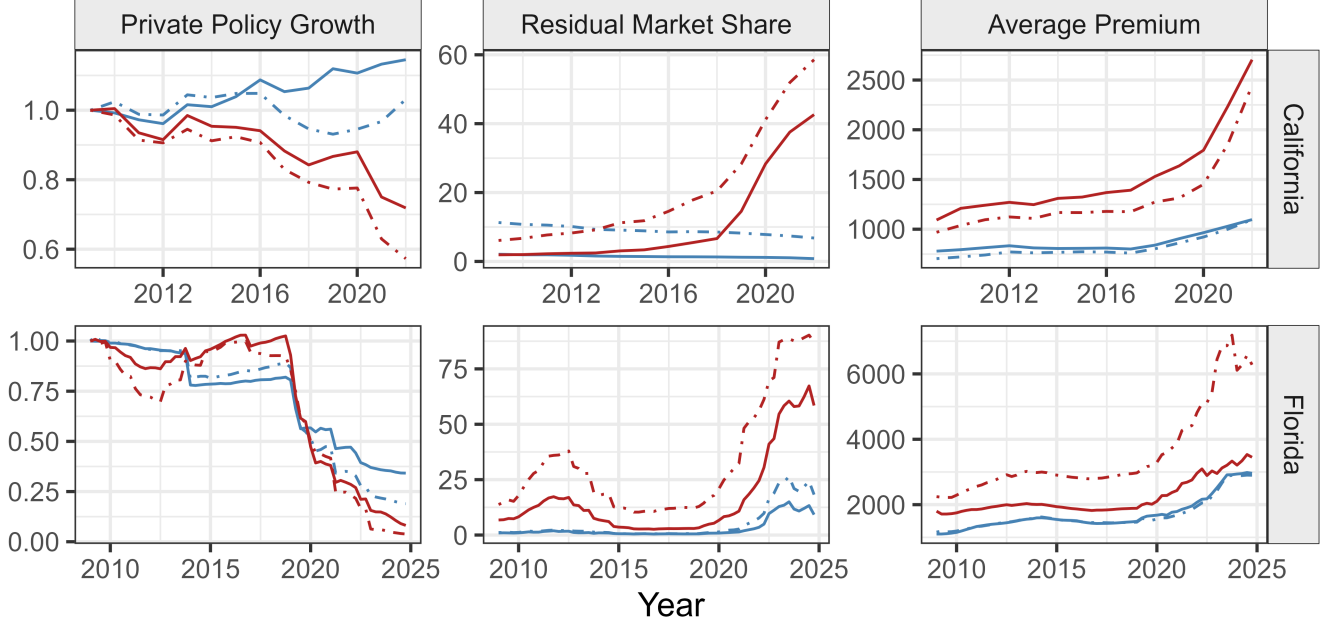


Figure 1. Insurance market trends in California (top row) and Florida (bottom row) in areas with differing disaster risk exposure. California (Florida) data is at the zip-code (county) level. Blue and red lines denote regions in the lowest (Q1) and highest (Q5) disaster risk quintile, respectively. Dash-dotted (solid) lines denote regions in which incumbent insurers have exposure-weighted average capital ratios below (above) the median. Private Policy Growth is the policy count from private insurers divided by the policy count in 2009. Residual Market Share captures the share of policies sold by the state residual insurer. Average Premium captures the nominal premium for a homeowners insurance policy that is renewed annually.

5-8 perform the same regression with the log of non-renewals as the dependent variable. Instrumenting changes in insurer net worth and capital ratios with out-of-zip-code disaster losses reveals that financial constraints increase insurance contractions and drive rising insurance premia.

Fact 2: Insurance market contractions negatively affect house prices conditional on weather exposure.

To illustrate the economic relevance of under-insurance beyond a shift in risk exposure from insurers to households, I investigate how insurance contractions transmit into housing markets. To test how insurance market conditions affect housing outcomes, I run the following specification:

$$Y_{z,t} = \beta M_{z,t} + \gamma' X_{c,t} + \eta_z + \tau_t + \epsilon_{z,t}, \quad (1)$$

where $Y_{z,t}$ is a zip-code level economic indicator, including the natural logarithm of median or mean house prices. The variable $M_{z,t}$ captures local insurance market conditions. I consider two proxies for $M_{z,t}$: (i) the logarithm of average homeowner insurance prices and (ii) the logarithm of the number of insurer cancellations. I construct insurance prices in zip code z in year t as the weighted average insurance prices over insurers active in zip code z :

$$\text{Insurance price}_{z,t} = \sum_{i \in I} w_{i,z,t} \times \frac{\text{Premiums}_{i,z,t}}{\# \text{Annual exposures}_{i,z,t}} \quad (2)$$

where

$$w_{i,z,t} = \frac{\text{Premiums}_{i,z,t}}{\sum_{i \in I} \text{Premiums}_{i,z,t}}$$

denotes the market share insurer i in zip code z in year t . The insurance price captures the average annual cost of a homeowners' insurance policy. I construct the number of insurer cancellation by summing the cancellations across insurers in given zip-code. These cancellations are initiated by the insurer and can thus be considered as a supply-driven change to insurance penetration.

The zip-code level fixed-effects η_z accounts for time invariant unobserved heterogeneity in insurance conditions and economic outcomes, such as wildfire risk perceptions that could correlate with both insurance prices and house prices. Time fixed effect τ_t captures common shocks across zip-codes. The vector $X_{z,t}$ contains control variables. Following the literature ([Eastman et al., 2024](#)), I control for various time-varying economic and demographic characteristics that affect jointly affect house demand and supply, and insurance prices.

The set of county-level controls includes demographic attributes (population, population density, unemployment rate, college attainment, the proportion of Black and Hispanic populations, the percentage of people with health insurance), income attributes (median household income, the percentage of households below the poverty line), and housing market attributes (median housing value, median year of construction, the percentage of owner-occupied homes, the vacancy rate, and the percentage of owner-occupied homes with a mortgage). In addition, I control for disaster damages in the past 5 years.

The set of zip-code level controls include the average amount of fire coverage per policy, separately including the entire policy limit and the limit for the house only, the number of existing policies, the average deductible per policy and the logarithm of county-level disaster damages, the number of insurers, and zip-code level insurer market concentration measured by the Herfindahl index. These insurance variables account for households endogenous selection of contract features that correlate with insurance prices. In addition, the insurance

market indicators account for correlations between growing weather risk, market power and insurance pricing.

Identification strategy The ideal experiment to estimate the effects of insurance contractions on the economy would capture changes in insurers’ supply decisions that are unrelated to risk exposure and local insurance demand. A key empirical challenge in understanding how insurance contractions affect macro-economic outcomes is that insurance prices simultaneously reflect climate risk exposure, the cost of living and possibly historical damages that directly affected property values. Proxies for insurance contraction, including insurance prices and policy cancellations, could affect macro-economic outcomes through each of these channels, while I am primarily interested in understanding the effects of insurance contractions conditional on disaster exposure and disaster realizations. While one could correct for observable disaster damages and risk scores, non-linear relationships between disasters, insurance payouts and property values could lead to model misspecification. In addition, equilibrium insurance prices are driven by both supply and demand. Potentially, high insurance premia reflect historical high insurance demand through increased awareness about climate risk that correlates indirectly with households wealth, leading to biased estimates.

I design an empirical strategy that isolates insurer supply shocks by exploiting the geographical breakdown of insurers’ activities. I instrument average zip-code level insurance prices and policy cancellations by disaster shocks occurring in other counties weighted by the incumbent insurers’ out-of-zip-code exposure shares. The empirical strategy is based on the testable assumption that, disaster losses in one region that drive the insurer to become financially constrained or adopt a conservative pricing strategy could affect insurance prices and supply decisions in other regions as well.

I construct the instrument as the zip-code level weighted average of insurers’ exposure to out-of-county damages⁴:

$$Z_{z,t}^{\bar{D}} = \log \left(\sum_{i \in \mathcal{I}_z} \omega_{i,z,t-1} \sum_{\substack{\bar{c} \in \mathcal{C} \\ d_{c(z),\bar{c}} > \bar{D}}} \omega_{\bar{c},i,t-1}^s \text{WeatherLosses}_{\bar{c},t-1} \right) \quad (3)$$

where $\mathcal{I}_z$ is the set of insurers active in zip code z and $\mathcal{C}$ is the set of all counties. The function $c(z)$ maps a zip code z to its corresponding county. The instrument considers that

⁴ While zip-code damages are available through in CDI, these (i) reflect damages only by reporting insurers and therefore may have the wrong magnitude, (ii) are available from 2018 onward only, and (iii) capture only insured damages. The use of county-level disaster data overcomes these issues.

disaster damages in the past year affects current insurance prices. While [Oh et al. \(2021\)](#) provide evidence of cross-subsidizing behavior of insurers across states, this analysis focuses on possible cross-subsidization within a state. The instrument can be interpreted as a shock to insurers cross-subsidization behavior. Such behavior is not driven by possible regulatory frictions, but rather updates of fire risk models or financial frictions that drive the insurer to adopt more conservative pricing within the entire state.

As before, $\omega_{i,z,t-1}$ captures the market share of insurer i in zip code z , lagged by one year to avoid capturing the effect of contemporaneous disaster losses on market withdrawals. The weight $\omega_{\tilde{c},i,t-1}^s$ captures the proportion of total premiums an insurer earns from county $\tilde{c}$, defined as:

$$\omega_{\tilde{c},i,t-1}^s = \frac{\text{Premiums}_{i,\tilde{c},t-1}}{\sum_{\tilde{c}' \in \mathcal{C}} \text{Premiums}_{i,\tilde{c}',t-1}}. \quad (4)$$

The variable $\text{WeatherLosses}_{\tilde{c},t-1}$ captures disaster losses as reported by the New York Fed. While these damages may reflect various natural hazards, damages are skewed towards fire hazards in California and hurricanes in Florida.

The validity of the instrument relies on out of county weather losses affecting house prices only through insurance prices and not through local risk perceptions. While regional disaster incidents could affect out-of-county risk perceptions, as long as distant damages, through the insurer's portfolio, are not structurally related to local risk perceptions, the exclusion restriction should be satisfied. The year fixed effects would absorb any changes in aggregate wildfire risk beliefs that could correlate with distant disaster damages and state-wide house prices.

To mitigate concerns about spatial correlation in damage that violate the exclusion restriction, losses in counties within $\bar{D}$ miles of each other are excluded, where $d_{c,\tilde{c}}$ is the distance between county c and county $\tilde{c}$. I measure distances between counties based on the centroid of the county. In a small number of cases, one-zip code could be part of more than 1 county. In this case, I allocate the county damages to the zip-code proportional to the share of the population in the zip-code that lives in a given county. The first stage becomes:

$$Y_{z,t} = \beta^z Z_{z,t}^{\bar{D}} + \gamma' X_{z,t} + \eta_z + \tau_t + \epsilon_{z,t} \quad (5)$$

Figure 2 plots the first-stage coefficient for different distance cutoffs $\bar{D}$. Higher distance cutoffs $\bar{D}$ imply that progressively more distant damages are used to instrument zip-code level insurance prices. The Figure shows that the predictive power of the instrument for insurance cancellations is maximized for intermediate distances around 200 miles. When

$\bar{D}$ is small, nearby disasters are included in the instrument. Consequently, local damages that are included as a control and zip-code level fixed effects capture most variation so that the instrument does not yield additional predictive power. When $\bar{D}$ is large, only a small proportion of actual damages to which insurers have exposure is covered, equally reducing its predictive power. The finding that the instrument has predictive power for insurance cancellation after including zip-code fixed effects that absorb time invariant fire risk exposures, suggests that the instrument does not only reflect risk. Relatedly, Figure (B.2) shows that county-level damages have close to zero autocorrelation, suggesting that it is hard to predict where future damages will occur.

In contrast, the first-stage coefficient of the instrument for insurance prices remains small. First-stage F-tests indicate that the instrument does not pass the Yogo-Stock F-value threshold of 10 for most distance cutoffs. For insurance prices, I construct an alternative instrument based on insurers' pricing errors. The instrument Z^{Resid} captures variation in local insurance prices that reflect deviations from the insurers' pricing model that is trained on other zip-codes.

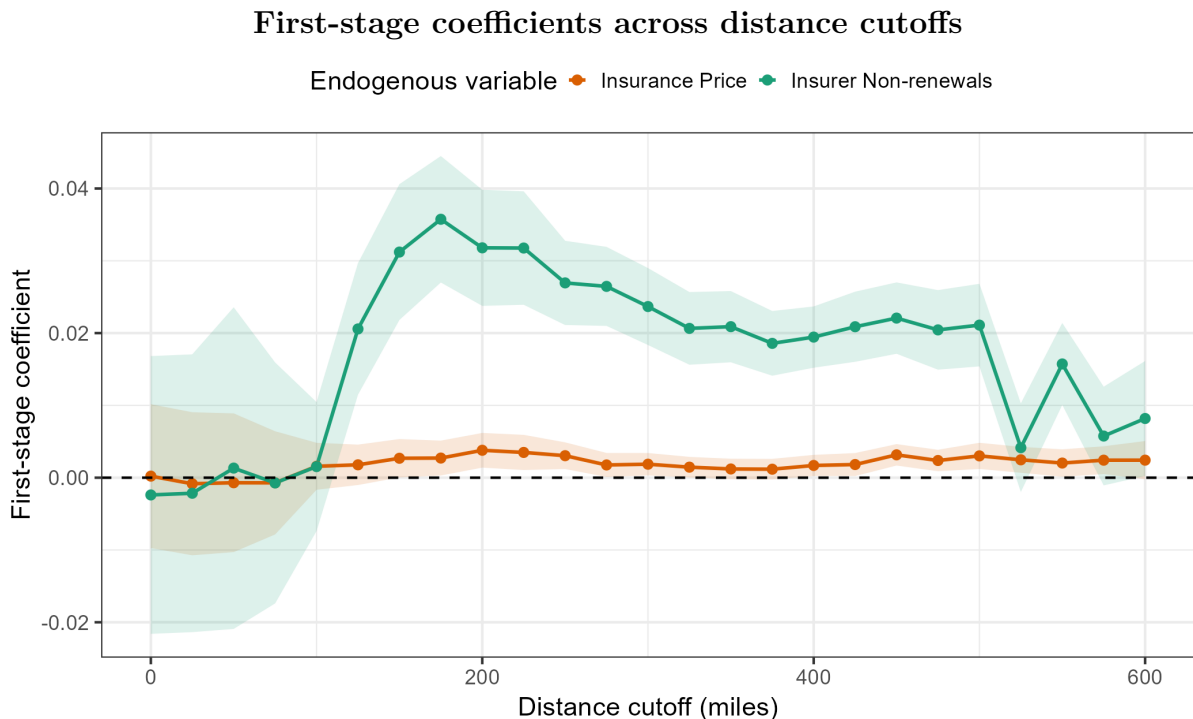


Figure 2. Figure reports the first stage coefficient of the instrumented on endogenous insurance variables. The instruments are constructed with different distance cutoffs $\bar{D}$, separately for the two instrumented variables insurance prices and insurer non-renewals. The zip-code level instrument captures the incumbent insurers' exposure to disaster losses in counties that are at least $\bar{D}$ miles away from the county that belongs to the zip-code, weighted by their zip-code market share.

Table 1 shows the first stage regression results for the instruments. The positive coefficients on log insurance renewals in column 2 shows that distant climate damages have affected local insurance conditions, revealing spatial spillovers. Columns 3 and 4 show the interaction with the average log of insurer net worth. The negative coefficient on the interaction term suggest that better insurer financial conditions reduce the volume on non-renewals following the same increase in disaster damages. Financial constraints thus affect the magnitude of spatial spillovers in insurance markets.

Table 1

First stage regression of zip-code level insurance indicators on the instrument, which reflects the logarithm of out of zip-code weather losses weighted by insurers' portfolio weights and local market shares. Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

	Log Ins. Price (1)	Log Ins. Non-renewals (2)	Log Ins. Price (3)	Log Ins. Non-renewals (4)	Δ Log RBC (avg) (5)	Δ Log Net Worth(avg) (6)
Z^{Resid}	0.63*** (0.09)		2.0*** (0.77)			
Z^{200}		0.03*** (0.004)		0.23** (0.09)	-0.01*** (0.001)	-0.003*** (0.0004)
Log Net Worth (avg)			0.06*** (0.01)	0.14 (0.09)		
$Z^{\text{Resid}} \times \text{Log Net Worth (avg)}$			-0.10* (0.06)			
$Z^{200} \times \text{Log Net Worth (avg)}$				-0.01** (0.006)		
Zip-code FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
F-test	52.2	51.7	153.3	81.9	38.5	27.1
Observations	14,469	8,029	14,469	8,029	14,935	15,046
R ²	0.92	0.96	0.93	0.96	0.29	0.71
Within R ²	0.23	0.05	0.29	0.05	0.02	0.02

Table 2 shows the second stage results of the regression of house prices on insurance indicators in both the OLS and IV specifications. The OLS specification shows that a 10% increase in insurance prices lowers median and average house prices by 0.8-1%, respectively. The IV specifications yield similar coefficients. Column 3 and 4 show that insurer non-renewals lower house prices. Finally, table A.4 shows that increases in the market share of the residual insurer similarly lower house prices. Overall, these results suggest that insurance contractions are costly for homeowners given their negative effect on house prices.

Table 2

Regression of house prices on insurance indicators in OLS- and IV specification. In the IV specifications, insurance prices are instrumented using insurer pricing errors weighted by their local market shares. Non-renewals are instrumented by the logarithm of out of zip-code weather losses weighted by insurers' portfolio weights and local market shares. Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

	Log(Median HP)		Log(Mean HP)		Log(Median HP)		Log(Mean HP)	
	OLS	IV	OLS	IV	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log Ins. Price	-0.08*** (0.03)	0.05 (0.05)	-0.10*** (0.02)	-0.08** (0.03)				
Log Non-renewals					-0.02*** (0.006)	-0.08** (0.04)	-0.002 (0.003)	-0.05*** (0.02)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip-code FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,768	12,731	11,753	11,746	7,361	7,361	7,002	7,002
R ²	0.96	0.96	0.99	0.99	0.98	0.98	0.99	0.99
Within R ²	0.08	0.07	0.16	0.15	0.04	0.04	0.17	0.17

Fact 3: The interaction of insurance and debt pins down house price fluctuations upon weather shocks.

The results thus far show that insurance prices and contractions can affect house prices conditional on disaster risk. Insurance conditions can also affect house price indirectly through disaster realizations. In the event of severe weather impacts, insurance can protect collateral values by enabling households to reconstruct their homes. In contrast, low insurance penetration can lower the supply of housing after disaster impacts relative to high insurance penetration and could increase house prices (Zivin et al., 2023). To test which explanation the data supports, I run the following local projection:

$$Y_{z,t+h} - Y_{z,t-1} = \beta_h \cdot \mathbf{1}_{z,t}^{fire} \cdot \text{High Insurance}_{z,t} + \gamma' X_{c,t} + \eta_z + \tau_t + \tau_t \times \tau_y + \varepsilon_{i,t+h} \quad (6)$$

where $\mathbf{1}_{z,t}^{fire}$ takes on value 1 if the zip-code experienced a severe wildfire. I define severe wildfires based on whether the zip-code is within or adjacent to a fire perimeter after a declared state of emergency is issued by the Governor of the state California. This data is available from 2019 onwards. $\text{High Insurance}_{z,t}$ equals one if the average amount of fire coverage per policy is higher than the median in a given year and zero otherwise. I include

$h \in \{0, 1, \dots, 6\}$ in the estimation. As before, $Y_{z,t}$ is the natural logarithm of median or mean house prices. I use the same set of controls as in regression (1) except I include 2 additional lags of the natural logarithm of median and mean house prices. I compute the response of house prices following a severe wildfire separately for the low- and high insurance group. Standard errors are clustered by zip-code.

Figure 3 shows that following wildfire incidents, house prices decrease significantly in zip-codes in which households have low fire insurance coverage. Specifically, average house prices drop by 3% by year 2 in zip-code that have the least fire coverage relative. In contrast, the house price drop in zip codes with high insurance coverage for following wildfire incidents is significantly less steep and recover more quickly. These heterogeneous effects show that insurance penetration plays an important role in how disaster shocks affect house prices. I do not find these effects for median house prices, which is consistent with the effects of wildfires on house prices concentrating among expensive houses.

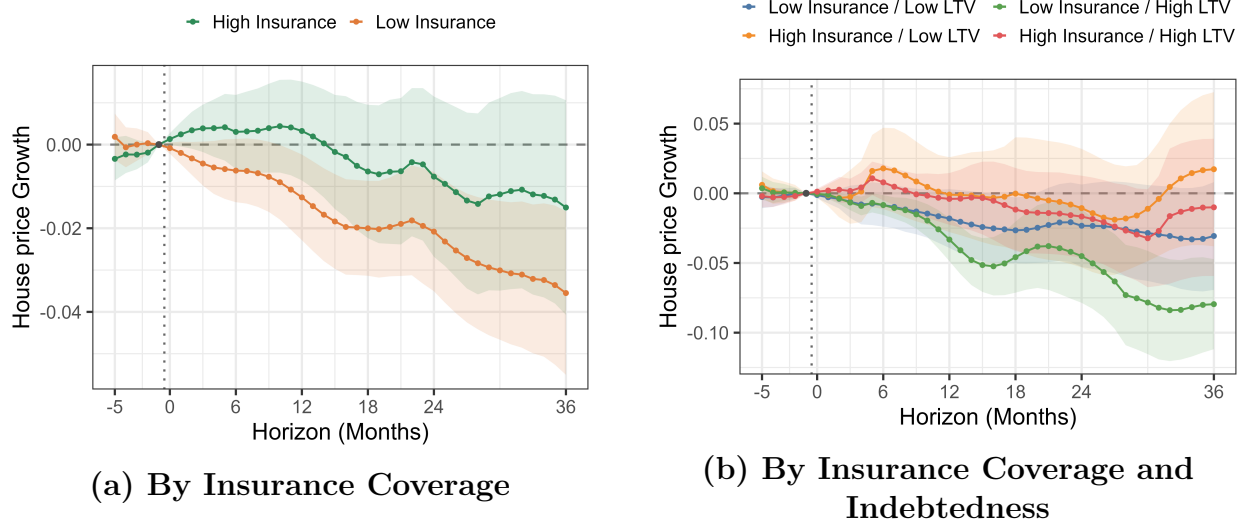


Figure 3. Percentage change in average zip-code house prices following wildfire incidents, by insurance coverage (panel a) and by insurance coverage and household indebtedness (panel b). Zip-codes are sorted into two insurance coverage groups based on whether their average fire coverage per policy is above or below the median. In panel (b), zip-codes within each insurance coverage group are further sorted into two indebtedness groups based on whether households' average loan-to-value ratio is above or below the median. Horizons are in months relative to the wildfire. Shaded areas reflect 95% confidence intervals. Standard errors are clustered at the zip-code level.

3 A dynamic housing model with insurance intermediaries

The empirical results show that insurance contractions are costly for homeowners by lowering house price. I develop a model of a small-open economy to study the interactions between insurance and housing markets in the presence of disaster risk. I first investigate how disaster shocks propagate to housing markets through insurance intermediaries. Second, I characterize the optimal insurance allocation that internalizes insurer fragility and the benefits of insurance to homeowners and compare the implicit tax with observed (implicit) subsidies.⁵

3.1 Disaster risk

Time is discrete and infinite. Disasters occur with probability p . Denote $\mathbf{1}_t$ as the indicator that equals 1 if a disaster occurred and 0 otherwise. Upon impact, a disaster destroys houses by a fraction ϕ , which I refer to as disaster severity. Weather risk θ_t is time-varying and follows an auto-regressive process:

$$\theta_t = \rho_\theta \theta_{t-1} + \varepsilon_{\theta,t} \quad \varepsilon_{\theta,t} \sim \mathcal{N}(0, \sigma_\theta^2) \quad (7)$$

I set $\phi = \frac{1}{1+\exp(-\theta)}$. The logistic transformation ensures that the fraction of destroyed houses is between 0 and 1. I discretize the process of weather risk using the Tauchen process.

3.2 Households

Households derive utility from consumption c_t , and housing s_t . Households can invest b_{t+1} into bonds to smooth consumption. In addition, households can invest into housing s_{t+1} , purchase insurance x_{t+1} to protect their housing assets and redraw home equity l_t . The representative household:

$$\max_{\{c_t, b_{t+1}, l_t, x_{t+1}, s_{t+1}\}} \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k u(c_t, s_t) \quad (8)$$

⁵The model is a variation of the frameworks proposed by [Bianchi and Mendoza \(2018\)](#) and [Jamilov and Monacelli \(2025\)](#). I modify [Bianchi and Mendoza \(2018\)](#)'s model by incorporating housing rather than capital that is subject to disaster shocks, and allow households to insure themselves in a private insurance market. I modify [Jamilov and Monacelli \(2025\)](#) by incorporating an intermediary in the form of an insurer rather than a bank.

where the utility takes the following form.

$$u(c_t, s_t) = \frac{1}{1 - \psi} (c_t + \nu_s g(s_t))^{1 - \psi} \quad (9)$$

The utility function thus features constant relative risk aversion (CRRA) over consumption and housing, comparable to the specification in [Iacoviello \(2005\)](#). Households choose next-period housing and current housing enters as a state variable in the household problem. Denote with ψ the coefficient of relative risk aversion. The parameter ν_s captures the utility weight on housing. The function $g(s_t) = \frac{s_t^{1 - \sigma_s}}{1 - \sigma_s}$ is increasing and concave in s_t . Utility is intertemporally separable but not intratemporally separable. Finally, the utility function satisfies the Inada conditions. The budget constraint reads

$$c_t + \frac{b_{t+1}}{R_t} + s_{t+1}p_t^s + p_t^x x_{t+1}s_{t+1} = b_t + w_t + s_t p_t^s - \mathbf{1}_t \phi_t p^r s_t + x_t \mathbf{1}_t \phi_t p^r s_t + l_t - f(l_t) + \Pi_t \quad (10)$$

where b_t denotes the stock of one-period bonds issued in the last period. R_t is the exogenous non state contingent world interest rate that satisfies $\beta R < 1$. Labor is normalized to 1 so that total labor income is w_t . Π_t is the sum of net transfers from insurers and profits from capital goods producers that are distributed lump sum to the household. The interest rate is stochastic and follows a stationary Markov process similar to disaster severity. While [Jamilov and Monacelli \(2025\)](#) endogenize the risk-free rate inside their a banking model, I follow [Bianchi and Mendoza \(2018\)](#) by assuming that the economy is small and open relative to world financial markets.

Housing s_{t+1} can be purchased at market price p_t^s . Housing assets are in fixed supply, so the market clearing condition is simply $s_t = 1$. Finding an equilibrium in the housing market involves finding a p_t^s such that, given household optimal decisions, the market clearing condition holds. The amount of housing purchased in the previous period that enters the budget constraint is s_t . I assume that following a disaster household rebuild immediately and incur rebuilding cost $\mathbf{1}_t \phi_t p^r s_t$, where p^r captures the cost to rebuild a house of size 1. The rebuilding cost p^r is assumed to be constant for simplicity. The total supply of housing thus remains constant.

Households can buy insurance x_{t+1} against next-period disaster risk from insurers. Insurance cover is defined as the fraction of possible housing that is destroyed in the next-period and can be bought at price p_t^x per unit of the exposed house.⁶ The insurance quantity

⁶Defining insurance cover as the fraction of possible future housing damage avoids modeling insurance payouts conditional on exceeding the insurance contract limit, which would preclude the differentiability of

bought gets aggregated across all insurers i , i.e. $x_t = \int_0^1 x_t(i) di$. The insurance payout based on insurance purchased in the previous period is conditional on a disaster having occurred and equals $x_t \mathbf{1}_t \phi_t p^r s_t$. The total insurance premium to cover next-period losses equals $p_t^x x_{t+1} s_{t+1}$ and is thus proportional to the housing investment, the fraction of the possible housing damage the household wishes to cover, the rebuilding cost p^r and the price per unit of insurance.

Households can redraw equity from their housing investments l_t , which comes at a cost $f(l_t)$ that captures transaction costs. The function $f(l_t)$ is strictly increasing in l_t , strictly convex and positive. Home equity redraws are small and play a modest role in the dynamics of the economy, but aid in the identification of the shadow value on the borrowing constraint.

Borrowing constraint The household can obtain an intertemporal loan $-b_{t+1}/R_t$ that involves interest payments. Borrowed funds can be diverted, which is avoided by imposing a collateral constraint. The collateral constraint limits total debt not to exceed a fraction κ_t of the market-value of beginning-of-period housing assets, which equals the post-disaster housing plus the insurance payout minus the home equity withdraw. The leverage constraint can thus be written as:

$$-\frac{b_{t+1}}{R_t} \leq \kappa_t (p_t^s s_t (1 - \mathbf{1}_t \phi_t) + x_t \mathbf{1}_t \phi_t p^r s_t - l_t) \quad (11)$$

The left-hand side equals total borrowing. The right-hand side equals housing collateral after the realization of any disaster shocks minus home equity redraws. Hence, κ_t puts a ceiling on the leverage ratio, but uses only post-disaster housing as collateral. This setup implies that disaster damages and insurance payouts affect collateral values and new borrowing capacity. Home equity redraws, by taking away collateral, further tighten the borrowing constraint.

Optimality conditions The representative household chooses $\{c_t, b_{t+1}, l_t, x_{t+1}, s_{t+1}\}$ to maximize their expected lifetime utility (8) subject to the budget constraint (10) and the borrowing constraint (11). In the baseline model, the household takes prices and insurer net worth as given. This maximization yields the following optimality conditions and constraints at each date t (see Appendix 10):

$$c_t :: \quad U'(t) = \lambda_t \quad (12)$$

the problem.

$$b_{t+1} :: \quad U'(t) = \beta R_t \mathbb{E}_t [U'(t+1)] + \mu_t \quad (13)$$

$$l_t :: \quad 1 = f'(l_t) + \frac{\mu_t \kappa_t}{\lambda_t} \quad (14)$$

$$x_{t+1} :: \quad p_t^x = \mathbb{E}_t \left[m_{t+1} \mathbf{1}_{t+1} \phi_{t+1} + \frac{\mu_{t+1} \kappa_{t+1} \mathbf{1}_{t+1} \phi_{t+1}}{\lambda_t} \right] \quad (15)$$

$$s_{t+1} :: \quad p_t^s + p_t^x x_{t+1} = \mathbb{E}_t \left[m_{t+1} \left\{ \nu_s g'(s_{t+1}) + p_{t+1}^s + x_t \mathbf{1}_t \phi_t p^r \right\} + \frac{\mu_{t+1} \kappa_{t+1}}{\lambda_t} Q_{t+1} \right] \quad (16)$$

$$c_t + \frac{b_{t+1}}{R_t} + s_{t+1} p_t^s + p_t^x x_{t+1} s_{t+1} = b_t + w_t + s_t p_t^s + (x_t - 1) \mathbf{1}_t \phi_t p^r + l_t - f(l_t) + \Pi_t \quad (17)$$

where $Q_{t+1} = p_{t+1}^s (1 - \mathbf{1}_{t+1} \phi_{t+1}) + x_t \mathbf{1}_t \phi_t p^r$ is the post-disaster value of the house plus the insurance payout, $\lambda_t = \mathcal{X}_t^{-\psi} = U'(t)$ is the marginal utility of consumption given the composite $\mathcal{X}_t = c_t + \nu_s g(s_t)$ and $m_{t+1} = \beta \lambda_{t+1} / \lambda_t$ is the stochastic discount factor.

Equation (12) is the first-order of the lagrangian with respect to consumption. Equation (13) is the Euler equation with μ_t denoting the multiplier on the borrowing constraint. It states the marginal benefit of borrowing exceeds the marginal cost by the value of relaxing the borrowing constraint μ_t . Condition (14) states that the marginal benefit of redrawing home equity should be equal to its marginal cost plus a financing cost when the collateral constraint becomes binding. The state contingent insurance price in equation (15) captures the cost to cover the expected damage. The insurance price equation sets the demand for insurance by equalizing the marginal benefit and the marginal cost. The marginal benefit equals the expected next-period's damage, adjusted for expected collateral constraints, multiplied by the stochastic discount factor m_{t+1} . If disasters lead to binding constraints in the next-period in expectation, households are willing to pay more for insurance. A binding collateral constraint implies selling houses at fire sale prices to repay debt which is costly. The fact that insurance helps prevent hitting binding collateral constraints in the next period makes that the expectation over μ_{t+1} ends up in the insurance valuation. The insurance price is furthermore convex in the disaster severity state θ . A high disaster severity today, given the AR(1) process, leads to allocating more weight to a high disaster severity states in the future as well, increasing the expected damage and therefore the insurance price.

The housing condition (16) equates the marginal cost of investing in housing today with the marginal benefit of housing in the next period. The marginal cost is the current house price p_t^s and the cost of insuring the house. High insurance premia today thus reduce house prices. The marginal benefit is the sum of the utility gain and capital gains, with the latter being the housing investment valued at next-period price p_{t+1}^s adjusted for any

disaster damages, and insurance payouts. Whenever $\mu_{t+1} > 0$ the collateral constraint binds in the next period and the fact that post-disaster housing serves as collateral increases the marginal benefit by $\kappa_{t+1} (p_{t+1}^s(1 - \mathbf{1}_{t+1}\phi_{t+1}) + x_t \mathbf{1}_t \phi_t p^r s_t - l_t)$.

Define $q_t^s = p_t^s + p_t^x x_{t+1}$ as the price of housing today and $q_{t+1}^s = p_{t+1}^s + \mathbf{1}_{t+1}\phi_{t+1}x_{t+1}$ as the value of the per-unit housing investment after realization of tomorrow's shocks. One can define the gross return on housing assets as:

$$R_{t+1}^s = \frac{\nu_s g'(s_{t+1}) + q_{t+1}^s}{q_t^s} \quad (18)$$

which states that the return on housing assets represents the marginal utility of housing plus the value of the house tomorrow including insurance payouts divided by the investment into the house today including insurance costs. Housing utility affecting house prices is consistent with the large literature studying house prices as a function of housing amenities through hedonic regressions. If the collateral constraint does not bind, i.e., $\mu_t = 0$, then iterating on (16) and using the transversality condition, we can express the house price as the expected present value of housing amenities discounted with R_{t+1}^s (as shown in appendix 10):

$$q_t^s = \mathbb{E}_t \left[\sum_{j=0}^{\infty} \left(\prod_{i=0}^j \mathbb{E}_{t+i} R_{t+1+i}^s \right)^{-1} \nu_s g'(s_{t+j+1}) \right] \quad (19)$$

In a world without financing constraints, the value of the house net of insurance payments is given solely by the discounted future housing benefits. We can combine equations (13) and (16) to derive the expected excess return on housing $R^{hp} \equiv \mathbb{E}_t [R_{t+1}^s - R_t]$:

$$R_t^{hp} = \mathbb{E}_t [R_{t+1}^s - R_t] = \underbrace{\frac{\mu_t}{\lambda_t \mathbb{E}_t [m_{t+1}]}_{\text{Liquidity premium}}} - \underbrace{\frac{\mathbb{E}_t [m_{t+1} \Upsilon_{t+1}]}{\mathbb{E}_t [m_{t+1}]}}_{\text{Collateral effect}} - \underbrace{\frac{\text{Cov}_t(m_{t+1}, R_{t+1}^s)}{\mathbb{E}_t [m_{t+1}]}}_{\text{Risk premium}} \quad (20)$$

where

$$\Upsilon_{t+1} = \frac{\kappa_{t+1} \mu_{t+1} q_{t+1}^s}{\lambda_{t+1} q_t^s} \quad (21)$$

is the marginal value of relaxing the borrowing at time $t+1$. The liquidity premium increases the return on housing when the collateral constraint binds as housing assets are illiquid and therefore cannot be used for consumption. The collateral effect partially offsets the liquidity premium. Investing in housing eases collateral constraints tomorrow, which reduces possible fire sales, and therefore affects expected excess returns through Υ_{t+1} . Finally, households demand a risk premium when investing in housing. The covariance term tends to be negative,

that is, low housing returns coincide with states where marginal utility of consumption is high.

If the liquidity premium dominates, then equations (41) and (20) imply a fire-sale mechanism. When collateral constraints become binding, the expected excess return on housing increases, and house prices today decrease. This can be interpreted as households collectively needing to sell assets in order to repay debt, which lowers equilibrium house prices. This in turn lowering aggregate borrowing capacity in the economy. The binding collateral constraint can result from either over-borrowing as in [Bianchi and Mendoza \(2018\)](#) or under-insurance that leads to more collateral being destroyed upon a disaster.

Overall, insurance take-up and affordability affects collateral values in two ways. First, low insurance take-up can result in a higher share of households becoming credit constrained upon disaster impact as they don't receive disaster payouts through (10). Low insurance payouts tighten the leverage constraint upon a disaster realization and drive housing fire sales tomorrow. Second, high insurance premia lowers house price today through (16) and could make the leverage constraint bind today.

3.3 Dynamic household problem

Denote $\mathcal{S}_t = \{\mathbf{1}_t, \theta_t, R_t\}$ as the realization of shocks at date t . The realization of a disaster $\mathbf{1}_t$ even though it is i.i.d, is a state variable as it is needed to computation of today's wealth and housing collateral.⁷ The household state variables are the individual states (s_t, b_t, x_t) and the aggregate states $(B_t, X_t, n_t, \mathcal{S}_t)$. The two endogenous state variables for the household are borrowing B_t and insurance X_t as they both affect future collateral and therefore future house prices. To form expectation about future house prices, households have a conjectured pricing function $p^s(B_t, X_t, n_t, \mathcal{S}_t)$, $p^x(B_t, X_t, n_t, \mathcal{S}_t)$ and $w(B_t, X_t, n_t, \mathcal{S}_t)$, and a perceived law of motion for the aggregate bond position $B_{t+1} = \Gamma^B(B_t, X_t, n_t, \mathcal{S}_t)$ and aggregate insurance position $X_{t+1} = \Gamma^X(B_t, X_t, n_t, \mathcal{S}_t)$. I need to track insurer net worth n_t as insurance prices, and therefore equilibrium insurance take-up, varies with insurer financial conditions. Yet, from the household's perspective, insurer net worth is exogenous. Finally, aggregate housing stock is not a state variable given that it is in fixed supply. The household's recursive

⁷ Alternatively, one can track collateral values and drop $\mathbf{1}_t$ as a state-variable, but this would not change the number of state variables. Tracking collateral in addition requires more grid points.

optimization problem is:

$$V(s_t, b_t, x_t, n_t, \mathcal{S}_t, B_t, X_t) = \max_{c_t, s_{t+1}, l_t, x_{t+1}, b_{t+1}} \{u(c_t, h_t, s_t) + \beta \mathbb{E}_{\mathcal{S}_{t+1} | \mathcal{S}_t} V(s_{t+1}, b_{t+1}, x_{t+1}, n_{t+1}, \mathcal{S}_{t+1}, B_{t+1}, X_{t+1})\} \quad (22)$$

subject to

$$\begin{aligned} c_t + \frac{b_{t+1}}{R_t} + s_{t+1}p_t^s + p_t^x x_{t+1} s_{t+1} &= b_t + w_t + p_t^s s_t (1 - \mathbf{1}_t \phi_t) + \mathbf{1}_t \phi_t s_t x_t p^r + l_t - f(l_t) + \Pi_t \\ -\frac{b_{t+1}}{R_t} &\leq \kappa_t (p_t^s s_t (1 - \mathbf{1}_t \phi_t) + \mathbf{1}_t \phi_t s_t x_t p^r - l_t) \\ B_{t+1} &= \Gamma^B(B_t, X_t, N_t, \mathcal{S}_t) \\ X_{t+1} &= \Gamma^X(B_t, X_t, N_t, \mathcal{S}_t) \\ N_{t+1} &= \Gamma^N(B_t, X_t, N_t, \mathcal{S}_t) \end{aligned}$$

The solution of this problem contains the policy rules $\hat{c}_t(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, $\hat{l}_t(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, $\hat{s}_{t+1}(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, $\hat{x}_{t+1}(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, and $\hat{b}_{t+1}(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$. The decision rules for bonds and insurance induces an actual law of motion for aggregate bonds, insurance, and net worth, given by $\hat{b}_{t+1}(\mathbf{1}, B_t, x_t, n_t, \mathcal{S}_t, B_t, X_t)$, $\hat{x}_{t+1}(\mathbf{1}, B_t, x_t, n_t, \mathcal{S}_t, B_t, X_t)$, and, $\hat{n}_{t+1}(\mathbf{1}, B_t, x_t, n_t, \mathcal{S}_t, B_t, X_t)$, respectively. Finally, the equilibrium contains actual pricing functions $\hat{p}^s(n_t, \mathcal{S}_t, B_t, X_t)$, $\hat{p}^x(n_t, \mathcal{S}_t, B_t, X_t)$ and $\hat{w}(n_t, \mathcal{S}_t, B_t, X_t)$.

3.4 Insurers

Whether insurance is available at affordable prices depends on insurers' risk-bearing capacity. This section outlines how insurers supply disaster insurance in the economy. There is a continuum of insurers of measure one selling insurance x_{t+1} to households and investing the proceeds into capital producing firms. The insurer is risk-neutral, accumulates net worth n_t , and maximizes the present discounted value of their life-time utility, V_t . Insurers start with an initial net worth $n \in \mathbf{N} \subset \mathbb{R}_+$, and choose an asset and liability configuration. How much insurance insurers can sell is constrained by regulation.

At time t , insurers collect premiums from sold insurance that can be directly invested. Possible claims are settled at time $t + 1$. The balance sheet constraint that must hold at all

times is given by⁸:

$$a_t = n_t + p_t^x x_{t+1} s_{t+1} \quad (23)$$

Let R_t^k be the return on the insurers investment portfolio from period $t - 1$ to period t . The law of motion of insurer-level net worth, n_t , can be written as:

$$n_t = a_{t-1}(1 + R_t^k) - \mathbf{1}_t \phi_t x_t p^r - \zeta^1 x_t^{\zeta^2}$$

Net worth today is the sum of prior period assets that generated return $1 + R_t^k$ minus claim payouts and operational costs. Claim payouts occur only when a disaster occurred and are proportional to the insurance exposure chosen in the previous period and costs to rebuild p^r . The introduction of convex operational costs allow for matching decreasing marginal propensities to insure across the insurer net worth distribution.

Regulatory friction Similar to [Gertler and Kiyotaki \(2010\)](#) and [Jamilov and Monacelli \(2025\)](#), I consider that the intermediary is subject to an incentive constraint. In the economy with insurers, the constraint limits insurers' ability to raise funding in the insurance market for investment purposes. The regulation constrains the insurer by ensuring that the insurers is able to pay out claims in most cases and thus acts as a Value-at-risk constraint. Let $V_t(n; \theta_t)$ be the maximized value of V_t given investment and insurance decisions at the end of period t . The regulation imposes that the insurer value is at least a fraction ζ of its exposure:

$$V(n_t; \theta_t) \geq \zeta x_{t+1} p^r s_{t+1} \quad (24)$$

The regulatory constraint thus puts a cap on the amount of insurance the insurer can sell.

Dynamic insurer problem The relevant state variables are net worth n_t and time-varying disaster severity θ_t . Insurers choose how much insurance to sell to households and how much to invest in assets, taking insurance prices and expected asset returns as given. Insurers may exit in the next period with i.i.d. probability $1 - \sigma$. Given that insurers are financially constrained, they have the incentive to accumulate assets. Imposing exogenous exits discounts future gains more and accumulating net worth becomes less valuable. The

⁸ Selling insurance generates liabilities, but at policy inception premium income is booked as a unearned premium reserve, which directly offsets the premiums collected on the asset side. Selling insurance therefore generally has no contemporaneous effect on net worth.

dynamic problem for the insurer is:

$$V(n_t, \theta_t) = \max_{\{x_{t+1}, a_t\} \geq 0} \left\{ \beta \mathbb{E}_{\theta_{t+1}|\theta_t} \left[(1 - \sigma)n_{t+1} + \sigma V(n_{t+1}, \theta_{t+1}) \right] \right\} \quad (25)$$

subject to

$$\begin{aligned} n_t &= a_{t-1}(1 + R_t^k) - \mathbf{1}_t \phi_t x_t p^r - \zeta^1 x_t^{\zeta^2} \\ a_t &= n_t + p_t^x x_{t+1} s_{t+1} \\ \zeta x_{t+1} p^r &\leq V(n_t, \theta_t) \end{aligned}$$

Insurers exit the market exogenously with probability $1 - \sigma$. Within the period, selling insurance increases leads assets to equal current net worth plus the quantity of insurance sold times the price of insurance.

One can substitute out the balance sheet constraint and the law of motion for net worth to obtain a simplified problem:

$$V(n_t, \theta_t) = \max_{\{X'\} \geq 0} \left\{ \mathbb{E}_t \Omega \left[\left(p_t^x (1 + R_{t+1}^k) - \mathbf{1}_{t+1} \phi_{t+1} p^r \right) x_{t+1} + n_t (1 + R_{t+1}^k) - \zeta^1 x_{t+1}^{\zeta^2} \right] \right\} \quad (26)$$

where $\Omega = \beta \left(1 - \sigma + \sigma \frac{V(n_{t+1}, \theta_{t+1})}{n_{t+1}} \right)$ is the insurers' augmented discount factor. Note that $s_t = 1$ as per the market clearing condition to simplify notation.

Unconstrained region Taking the first-order conditions w.r.t. x_{t+1} yields:

$$\mathbb{E}_t \left[\Omega \left(p_t^x (1 + R_{t+1}^k) - \mathbf{1}_{t+1} \phi_{t+1} p^r - \zeta^1 \zeta^2 (x_{t+1}^{uncon})^{\zeta^2-1} \right) \right] = 0 \quad (27)$$

where x_{t+1}^{uncon} is the optimal insurance supply of the insurer when the regulatory constrained is assumed to be not binding. When the regulatory friction is not binding, insurers sell insurance up to the point where the marginal value of generating assets equals the marginal cost on the insurance portfolio. When investments are profitable, insurers seek to sell more insurance that generate investable premiums. This effect increases in the insurance price at which insurers can sell insurance. Even if premiums are below expected claim payouts, if selling insurance generates assets that can be invested at a higher yield, the insurer will choose to do so.

Define the concept marginal propensity to insure (MPI) as the elasticity of insurance supply with respect to insurer net worth. In the unconstrained region, insurance supply is

not dependent on net worth and $MPI^{uncon} = \frac{\partial x_{t+1}(n, \theta)}{\partial n_t}$ therefore equals zero.⁹

Constrained region When the constraint binds, one can substitute $\zeta x_{t+1} s_{t+1} = V(n_t, \theta_t)$ and derive the insurers' policy function for insurance $\mathcal{X}$ as an implicit function of choice variable x_{t+1} :

$$\mathcal{X}^{con}(n, \theta) = \frac{\mathbb{E}_t \left[\Omega \left(n_t (1 + R_{t+1}^k) - \zeta^1 x_{t+1}^{\zeta^2} \right) \right]}{\zeta p^r - \mathbb{E}_t \left[\Omega \left(p_t^x (1 + R_{t+1}^k) - \mathbf{1}_{t+1} \phi_{t+1} p^r \right) \right]} \quad (28)$$

As in the unconstrained region, insurance supply is increasing in expected asset returns and insurance prices, and decreasing in expected claims and operational costs. Insurance supply furthermore decreases in regulatory stringency ζ . In addition, insurance supply depends on net worth as net worth increases the value function V_t and therefore how severe capital constraints bind.

If regulatory constrained, insurers can only sell insurance up to the point where the regulatory constraint becomes binding, leaving a wedge between the marginal value of assets and the marginal cost on insurance. The spread between the return on the investment portfolio and return on the insurance portfolio should increase in the degree to which insurers face capital constraints. Figure 4 plots the average return on assets and return on insurance (ROI) in the U.S. combining all lines of business across all P&C insurers (left panel) and by RBC Capital ratio tercile (right panel). The ROI is computed as $100 * (1 - \text{Combined Ratio})$, where the Combined Ratio is the ratio of claims plus other operational costs divided by premiums collected. Over the past 25 years, on average, insurance operations generated a negative return of -1.2% on average, while the investment portfolio yielded 3.2% per year on average. The operational spread, defined as the difference between ROA and ROI, is higher for insurers that are more capital constrained, which is consistent with (??).

In the constrained region, MPI is:

$$MPI^{con} = \frac{\partial \mathcal{X}_{t+1}(n, \theta)}{\partial n_t} = \frac{\mathbb{E}_t \left[\Omega(1 + R_{t+1}^k) \right]}{\zeta p^r - \mathbb{E}_t \left[\Omega \left(p_t^x (1 + R_{t+1}^k) - \mathbf{1}_{t+1} \phi_{t+1} p^r \right) \right] + \zeta^1 \zeta^2 x_{t+1}^{\zeta^2-1}} \quad (29)$$

The equation shows that MPI is increasing in expected asset returns and insurance prices, but decreasing in regulatory stringency and operational costs. High ζ reflect stringent capital

⁹ Changes in net worth affect total assets in the economy and therefore affect expected asset returns. Yet, in equilibrium expected asset returns are taken as given for an infinitesimal insurer, yielding zero MPI in the unconstrained region.

U.S. P&C insurers - Return on Assets and Return on Insurance

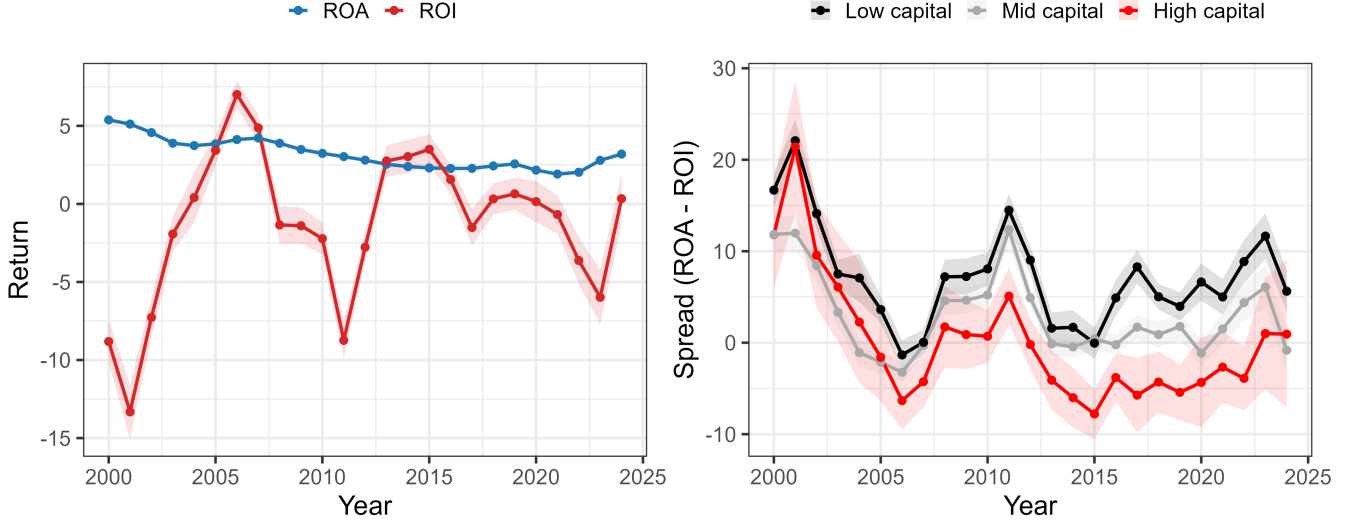


Figure 4. Return on Assets (ROA) and Return on Insurance (ROI) across all U.S. P&C insurers in all lines of business. The left panel shows the equal weight average of insurers' ROA and ROI by year. The right panel shows the spread, defined as the difference between ROA and ROI, by year and RBC ratio tercile.

regulation and indicates that for a given change in net worth, insurance supply contractions are small.

3.5 Final good firms

There is a continuum of measure one of perfectly competitive firms that produce the final good, y_t . Final good use an identical constant returns to scale Cobb-Douglas production function with aggregate capital, k_t , and labor, h_t , as inputs:

$$y_t = z_t k_t^\alpha h_t^{1-\alpha}, \quad 0 < \alpha < 1 \quad (30)$$

where z_t is aggregate productivity. Final good firms produce, pay wages to households and generate returns on assets for insurers. z_t follow a Markov process. Capital of final good firms is not destroyed by disasters in the baseline model.

The capital-producing firm is cash-strapped and depend on insurers and external investors for external financing. Total assets in the economy equal:

$$\bar{a}_t = a_t^{out} + a_t^{ins} \quad (31)$$

where a_t^{out} equals assets from outside investors and a_t^{ins} equals assets coming from insurance companies.¹⁰ Each capital-producing firm takes in claims $\bar{a}_t(j)$ and produces $\Phi(\bar{a}_t(j))$ new units of the capital good, where $\Phi(\cdot)$ is a transformation function such that $\Phi' > 0$ and $\Phi'' < 0$. The decision problem of each producer is thus:

$$\max_{\bar{a}_t(j)} q_t^{\bar{a}} \Phi(\bar{a}_t(j)) - \bar{a}_t(j) \quad (32)$$

Given symmetry across capital producers ($\bar{a}_t(j) = \bar{a}_t$), the aggregate price of capital is pinned down as follows:

$$q_t^{\bar{a}} = [\Phi'(\bar{a}_t)]^{-1} \quad (33)$$

Profits from capital goods producers are redistributed lump sum to households. Finally, the return on capital that insurers and outside investors take as given is given by:

$$R_{t+1}^k = \frac{z_{t+1} \alpha k_{t+1}^{\alpha-1} h_{t+1}^{1-\alpha}}{q_t^{\bar{a}}} \quad (34)$$

After being used in the production of the final good, capital fully depreciates.

4 Unregulated Decentralized Competitive Equilibrium

A recursive competitive equilibrium consists of the insurer value and policy functions, household policy functions, such that, given a vector of aggregate pricing $\{p^s, p^x, w, R^k\}$ functions and the perceived law of motions for savings $\Gamma^B(B_t, X_t, n_t, \mathcal{S}_t)$, insurance $\Gamma^X(B_t, X_t, n_t, \mathcal{S}_t)$, and net worth $\Gamma^N(B_t, X_t, n_t, \mathcal{S}_t)$:

1. The value and policy functions of households and insurers solve their corresponding decision problems.
2. The perceived law of motion for aggregate bonds and aggregate insurance, and perceived pricing functions are consistent with the actual decisions rules and actual prices, respectively.
3. The goods market clears: $c_t = y_t + b_t - b_{t+1}/R_t - p_t^s \mathbf{1}_t \phi_t$ ¹¹, the insurance market clears: $x_{t+1} = \psi_t n_t$, the housing market clears: $s_{t+1} = 1$ and the asset market clears:

¹⁰ Considering that insurers offer insurance to cover houses of size $s = 1$, their size in the economy is modest. Introducing outside investors ensures that the fraction of equity held by insurers in the economy matches reasonable macro-economic averages.

¹¹ Consumption thus equals total production net of foreign capital flows and destroyed housing capital.

$$k_{t+1} = \Phi(a_t)$$

4. The household resource, collateral and insurer balance sheet constraints hold.

5 Taking the Model to the Data

The section calibrates the model. I first motivate the choice of parameters and then discuss the model’s performance in matching empirical moments.

5.1 Parameterization

Table 3 shows the parameter values used in the baseline model, which I motivate next.

Disaster parameters I calculate the disaster probability p as the fraction of zip-codes that experienced positive catastrophic fire damages in the state California during 2018-2023 (excluding indirect smoke damages from neighboring zip-codes), yielding an estimate of 0.0523. I compute the parameters ρ_θ and σ_θ based on county- and state-level disaster damages. Figure B.2 plots the distribution of auto-correlation coefficients in annual inflation-adjusted disaster damages at both state and county-level. The state- and county-level median autocorrelation are both near 0, although there exists a substantial number of geographies with either strongly negative or strongly positive autocorrelation. The median volatility of disaster damages is 0.9 at both the state and county level. These statistics suggest that the size of disaster damages is largely random and show little persistence.

Household parameters I choose a simple equity redraw cost function $f(l_t) = \frac{\theta^l}{2}l^2$ and set θ^l so that average home equity redraws as a share of home values equal 14.1% (Freddie Mac, 2022). I set ν_s equal to match average housing costs as a share of total household spending of 0.33 (U.S. Bureau of Labor Statistics, 2025). Lastly, I set rebuilding cost p^r to match the ratio of rebuilding costs to the market value of the house upon a disaster of 62.5% (Costa and Baker, 2024).¹²

Insurer parameters I set $\bar{a}^{out}$ to match the market share of P&C insurers in the corporate bond market of 3.3%, which is derived using the ownership data reported by Ge and Weisbach (2021). I set ζ to match the fraction of insurers that are considered capital constrained. I

¹² (Costa and Baker, 2024) estimate that following Californian wildfires replacements costs as a fraction of home values range between 45% and 80%.

Table 3
Model Parametrization

Parameter	Value	Description	Target/Source
<i>Households</i>			
ψ	2	Risk aversion	Literature
β	0.97	Discount factor	Literature
ν_s	0.038	Weight on housing utility	Housing to total exp. 0.33
σ_s	0.9	Housing utility concavity	-
$\{\kappa^H, \kappa^L\}$	$\{0.95, 0.75\}$	Collateral constraint states	Average LTV
p^r	0.76	Housing rebuilding cost	Rebuilding cost / p^s upon shock 0.625
θ^l	3.6	Home equity redraw cost l	l / p^s 0.14
<i>Firms and Technology</i>			
α	0.33	Average Capital share	Literature
$\{z^L, z^H\}$	$\{0.994, 1.006\}$	Aggregate productivity states	Output volatility 0.01
a^{out}	26	Capital outside investors	3.3% share P&C ins. in Corp. Bonds
$\bar{a}$	5.2	Capital production parameter	Average investment yield 3.2%
<i>Insurers</i>			
σ	0.90	Insurer survival rate	Observed exit rate
ζ	0.98	Capital constraints parameter	Share constrained insurers 7.9%
ξ	0.073	Size entering insurers	Insurance gap of 45%
$\bar{\zeta}_1$	0.033	Non-claim expenses, linear	Average MPI 0.5
$\bar{\zeta}_2$	2	Non-claim expenses, quadratic	Jamilov and Monacelli (2025)
<i>Stochastic Processes</i>			
γ	0.053	Disaster probability	Estimated
ρ_θ	0.0	Persistence, disaster severity	Estimated
σ_θ	0.9	Volatility, disaster severity	Estimated
π_z^{LL}, π_z^{HH}	0.9, 0.9	Productivity persistence	Literature
$\pi_\kappa^{HL}, \pi_\kappa^{LL}$	0.1, 0.0	Collateral-regime transitions	Literature
ρ_R	0.68	Persistence, Interest rate	Literature
σ_ς	1.86%	Volatility, Interest rate	Literature
$\bar{R}$	2.0%	Mean, Interest rate	Literature

empirically estimate the fraction of active insurers who have RBC ratios below 300, which is the point at which insurers start receiving regulatory scrutiny. I set ξ_t to match the insurance gap of 45% in the U.S. ([Aon, 2026](#)).¹³ I set ζ^1 to match empirical estimates of

¹³The insurance gap in the U.S. varies over time. Exhibit 16 in [Aon \(2026\)](#) shows that the average equals roughly 45%.

marginal propensity to insure. I estimate MPI through a simple OLS regression:

$$\Delta X_{izt} = \beta \Delta n_{it} + \phi_i + \eta_z \times \tau_t + \epsilon_{i,t}, \quad (35)$$

where $\Delta X_{izt} = \log X_{izt} - \log X_{izt-1}$ are insurer zip code level changes in sold exposures and $\Delta n_{it} = \log n_{it} - \log n_{it-1}$ captures changes in net worth. Zip-code by Year Fixed effects $\eta_z \times \tau_t$ capture local insurance shocks that affect insurers operating in this zip code. I therefore estimate how changes in net worth affect insurance quantities by comparing insurers within a zip-code year pair, i.e., by keeping local insurance conditions constant. Insurer fixed effects ϕ_i absorb time-invariant insurer traits, such as risk-taking, that could affect growth or shrinkage in the insurer's portfolio.

Equilibrium insurance quantities and net worth are jointly determined and an OLS estimate of β would be subject to a simultaneity bias. However, zip-code level insurance revenues are small compared to overall insurer net worth. Hence, changes in zip-code level revenues are unlikely to individually materially affect insurer balance sheet conditions, which alleviates the simultaneity bias.

Panel A in Table A.1 shows that changes in insurer net worth are an important determinant of local insurance quantities. Empirical estimates of MPI range between 0.27 and 0.75. Panel B shows similar findings for insurer exit decisions: insurers with higher net worth levels and increases are less likely to exit zip-codes. I assume an average marginal propensity of 0.5 to match in the quantitative exercise.

Table A.2 investigates heterogeneity in MPI across insurers. Columns 1 and 2 show that insurers with low capital ratios and highly leveraged insurers have high MPI. This is consistent with insurers who are operating close to binding capital constraints experiencing larger volatility in their supply following changes in net worth. Column 3 indicates that insurers who have higher market shares tend to have lower MPI on average. The supply of insurers who enjoy local market power is thus less sensitive to net worth fluctuations.

Production parameters Aggregate productivity z_t follows a Markov chain with two states $\{z_l, z_h\}$ and corresponding transition probabilities $\{\pi_{LL}, \pi_{HH}\}$. The global interest rate R follow an independent AR(1) process. The AR(1) process for the logged gross real interest rate is $\ln(R_t) = (1 - \rho_R)\bar{R} + \rho_R \ln(R_{t-1}) + \varsigma_t$ with $\varsigma_t \sim \mathcal{N}(0, \sigma_\varsigma)$. The interest rate is discretized using Tauchen's quadrature method with three realizations.

Following Jamilov and Monacelli (2025), I assume that the elasticity of output with respect to capital is $\alpha = 0.33$ and that the capital production function takes the following

form:

$$\Phi(\bar{a}) = a\bar{a}^b, \quad a, b > 0 \quad (36)$$

Parameter b is set to 0.75 so that the elasticity of the price of capital with respect to equity investments is equal to 0.25. I calibrate a so that the average return on capital R^k in the baseline simulations is 3.2%.

5.2 Targeted and Untargeted moments

Table 4 presents the calibration results. The model matches most targeted moments, although the model predicts a larger share of insurers that are constrained. The model matches untargeted moments reasonably well, including housing damages as a fraction of GDP and the ROA-ROI spread on insurers' portfolios.

Table 4
Model Averages and Calibration Targets in the baseline economy

	Model	Target / Data
<i>Targeted moments</i>		
Insurance gap	49%	45%
Return on invested assets	3.5%	3.2%
Share insurers in corporate bonds	3.2%	3.3%
Marginal Propensity to insure	0.49	0.50
Share constrained insurers	37.0%	27.6%
Housing cost to total expenditures	0.34	0.33
Rebuilding cost to home value upon shock	0.63	0.63
Home equity redraw to home value	0.15	0.14
Output volatility σ_y	0.002	0.01
<i>Untargeted moments</i>		
Housing damages to GDP	0.68%	0.60%
ROA-ROI spread	0.73%	3.0%
Premium income to net worth	2.4%	3.8%

6 Baseline Model Results

This section first presents the policy functions of the household and insurer. It then shows the Impulse Response Functions after a disaster shock under different levels of insurer net worth.

6.1 Policy functions

Household policy functions Figure 5 shows the policy function for the household as a function of the savings state b , separately for the no-disaster and disaster case. The disaster case is split by a low- and high insurance case. Consumption is higher in the no-disaster case than in the disaster case. Upon a disaster, housing collateral gets destroyed. Given that households buy houses of size 1 each period, fewer resource are available for consumption. The drop in consumption is larger for the low-insurance case, highlighting the importance of insurance in preserving economic activity. Furthermore, the slope between consumption and savings is steeper in the disaster than in the no-disaster case. The non-linear relationship between consumption and savings in the disaster states reflect the housing fire sales dynamic. If households have high debt (negative savings), house prices drop more upon a disaster shock compared to when households have low debt. The steeper slope ‘kicks in’ earlier for low levels of insurance, highlighting the important interactions between disasters, debt and insurance in determining consumption.

Next-period’s savings choice is an increasing linear function of current savings without a disaster. In contrast, disaster destroy collateral and erode borrowing capacity, especially in states of high debt (negative savings) through the same fire sale dynamics. Hence, for sufficiently low levels of savings, the slope between current savings and next-period savings turns negative. For low levels of savings today, next-period savings are higher for the high insurance case, given the lower borrowing capacity relative to the high insurance case.

Next-period insurance choice decreases in savings. Both savings and insurance act consumption smoothing devices. As insurance and savings can be viewed as substitutes, when savings are high, households’ need to buy insurance reduces. The change in the new insurance choice upon a disaster mimics the borrowing choice. A low-insurance state with a disaster destroys borrowing capacity and households have less resources to buy new insurance for the next period. Under-insurance is therefore persistent in the short-term.

Finally, the collateral value Q substantially drops in the disaster case and is by construction lower in the low-insurance case. Using the high-insurance equilibrium, I compute a low-insurance counterfactual by lowering the collateral value by the difference in insurance payouts between the high- and low insurance state. Given that the computed counterfactual relies on houses prices in the high-insurance equilibrium, the difference between the counterfactual and low-insurance collateral value reflects the fire sale effect. The fire sale effect is close to zero in low debt states, but becomes more positive for high debt states.

Household policy functions

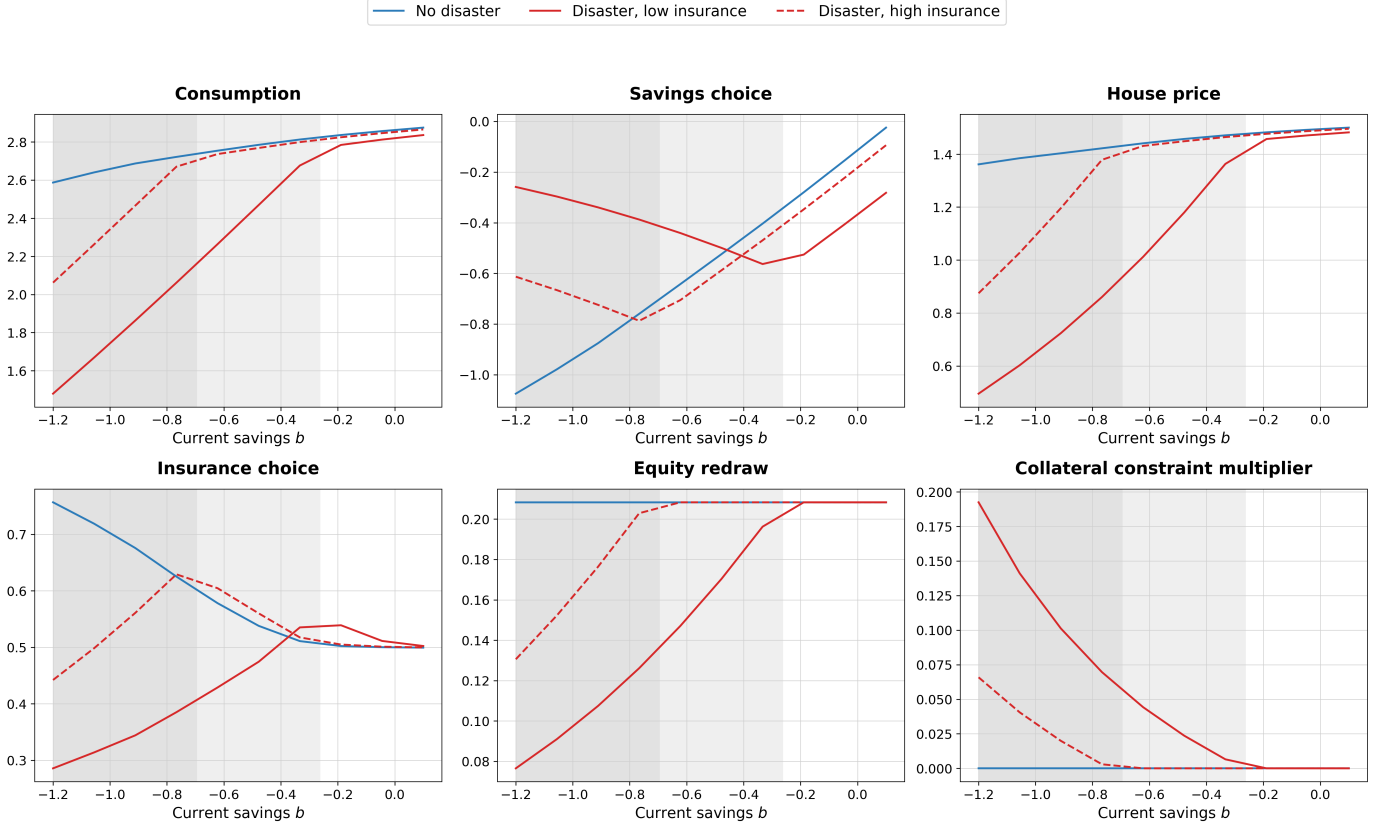


Figure 5. Household consumption, savings, house prices, insurance, equity redraws and the collateral constraint multiplier as function of outstanding savings in the decentralized competitive equilibrium. The blue line corresponds to the state where no disaster occurred. The red solid (dashed) line corresponds to a disaster state where the household insurance state is the highest (lowest) on the grid. κ is set to its highest value. All other states are set to their median. The light (dark) shaded area represents the area in which the household with low (high) faces binding collateral constraints.

Insurer policy functions Figure 6 plots equilibrium insurance supply, insurance prices, insurer leverage and marginal propensity to insure as a function of the insurer net worth state, and separately for the disaster- and no-disaster state. The shaded area reflects the area in which insurers are facing binding capital constraints.

There are several observations to make. First, in the constrained region, insurance supply sharply rises in net worth, whereas it is independent of net worth in the unconstrained region, which reflects the features of marginal propensity to insure discussed in section 3.4. The only variation of insurance supply across net worth in the unconstrained region happens indirectly through equilibrium insurance prices and expected returns. Second, insurance prices are a decreasing function of insurer net worth. Insurance prices are high when insurers have low net worth and are constrained, consistent with the empirical evidence presented in Table A.3. Third, MPI is high when insurer net worth is low, which is consistent with equation 29

that shows that that MPI is decreasing in the size of the insurance portfolio x_{t+1} . Fourth, expected asset returns decrease in insurer net worth. Higher insurer net worth increases total capital in the economy and depresses returns.

An important feature of the insurer's equilibrium behavior is the sharp reduction in insurance upon binding capital constraints, which can be costly for homeowners as they face a reduction in the availability of insurance. Section 7 discusses how a social planner can appropriately tax or subsidize insurance to limit the frequency of visiting these states.

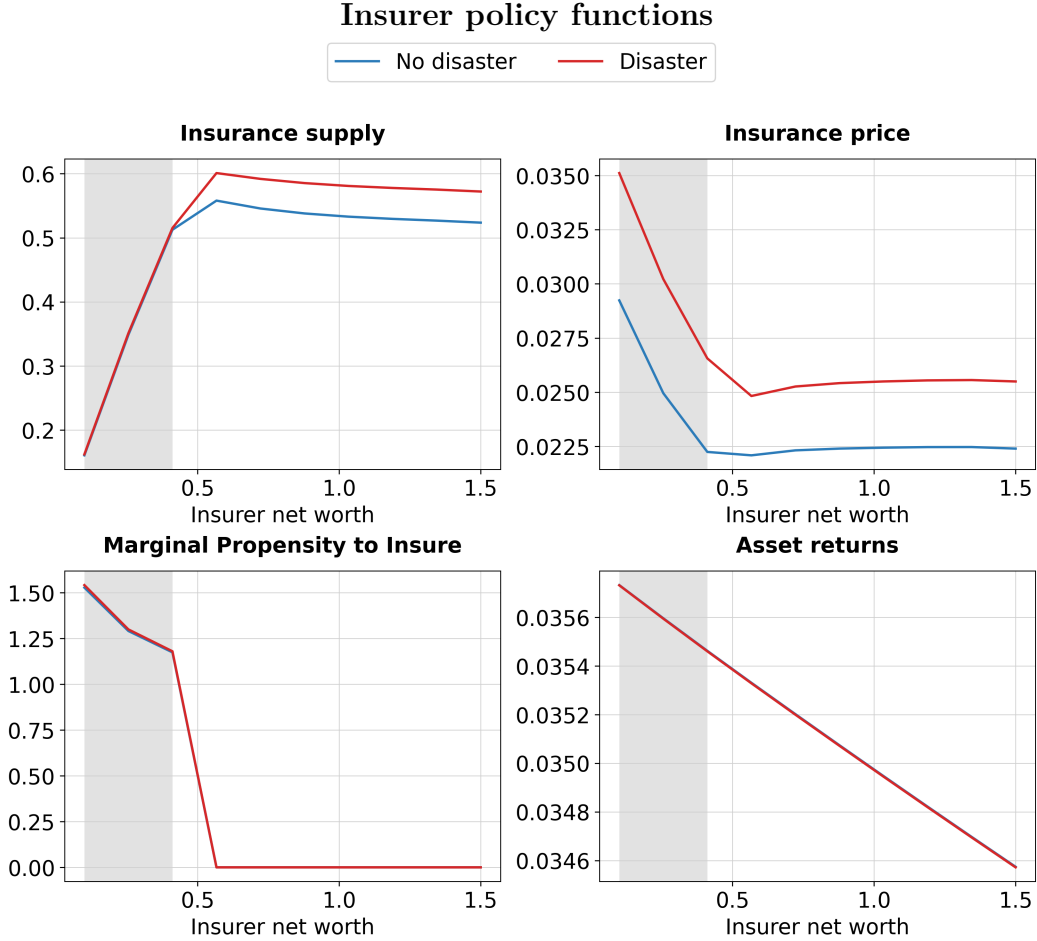


Figure 6. Insurer assets, insurance supply and insurance prices as a function of insurer net worth. The blue line corresponds to the state where no disaster occurred. The red solid (dashed) line corresponds to a disaster state where the household insurance state is the highest (lowest) on the grid. All other states are set to their median.

Transmission of insurance market conditions into house prices Figure 7 highlights how insurance market conditions affect welfare and house prices. The left-panel shows that household welfare is increasing in net worth, consistently across different levels of savings.

Insurer net worth is beneficial for households as it keeps insurance affordable and through more insurance take-up prevents future sharp reductions in consumption. The right panel shows that house prices increases in insurer net worth, and the shape mimics that of the relationship between insurance prices and net worth (see Figure 6). House prices increase in insurer net worth through the direct effect of insurance premiums on marginal utility, affecting house valuations, but also through the expectation of future house prices that depends on future insurance coverage.

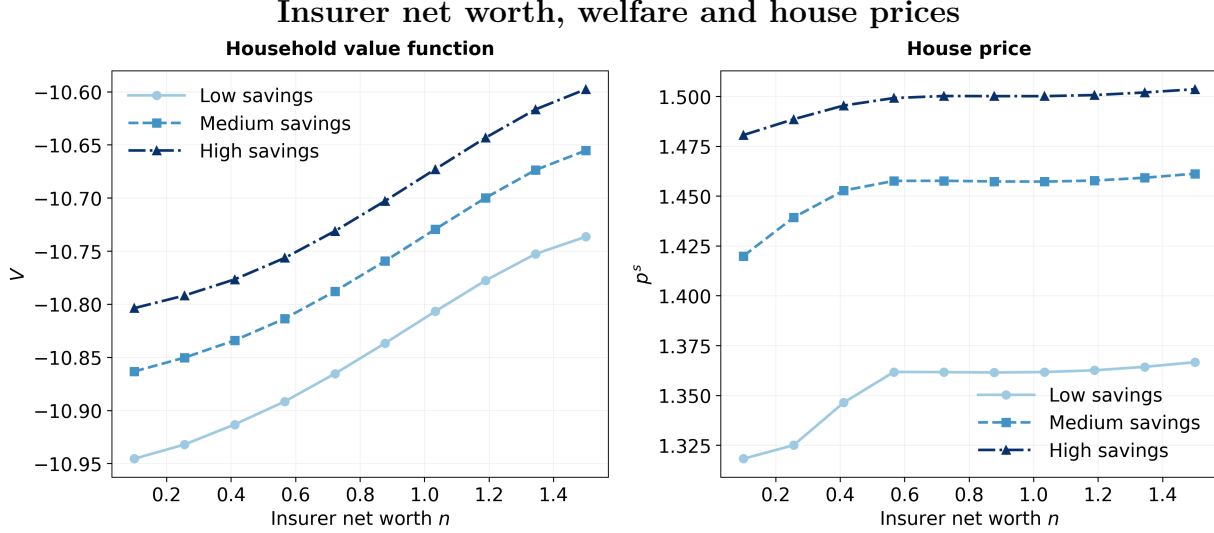


Figure 7. The household value function in the decentralized equilibrium and house prices as a function of insurer net worth for different states of savings.

I next exploit the variation of insurance and house prices across the net worth grid to estimate the elasticity of house prices with respect to insurance prices. Keeping other states fixed, I obtain a mean elasticity of -0.13 and a median elasticity of -0.10, which aligns with the empirical estimates in Table 2.

6.2 Impulse response functions

I study impulse response functions (IRFs) to a disaster shock under different levels of insurer net worth. To obtain these IRFs, I first run a simulation for $T=500$ and discard a fraction off earlier periods. At time $T^* < T$ as represented by the dashed vertical lines, a disaster shock occurs (timing random). I run a second simulation that is identical to the first except that there is no disaster shock. The impulse response is constructing by subtracting the second from the first simulation. In the fragile economy simulation, a negative financial shock to insurers occurs at $T^* - 1$, which reduces their net worth and therefore equilibrium

insurance quantities. The fragile economy reflects the possible scenario in which insurers are recurrently hit by disasters or broader economic shocks, shrinking their risk-bearing capacity, or when uncertainty is high so that effective risk-taking by insurers lowers. In the baseline simulation, the shock to net worth is absent. Comparing the baseline and fragile simulation informs how the state of the insurance sector affect the propagation of disaster shocks on economic outcomes.

Figure 8 presents the results. A disaster shock significantly reduces consumption and house prices. The drop is steeper in the fragile economy, revealing that the state of the insurance sector is important in the propagation of natural disasters to the economy. In the baseline economy, when equilibrium insurance take-up is relatively high, savings drop as households want to smooth consumption. In contrast, in the fragile economy, relatively low insurance erodes borrowing capacity upon a disaster shock, and households' ability to smooth consumption weakens.

Insurance supply drops upon the disaster shock, but is accompanied by either increasing or decreasing insurance prices. In the baseline economy, equilibrium insurance prices increase upon the shock. As net worth reduces and equilibrium insurance take-up drops, the likelihood of the collateral becoming binding in the future increases, increasing the insurance price (see equation (15)). In the fragile economy, marginal utility of consumption increases more severely as the drop in consumption is larger, depressing the willingness to pay for insurance coverage.

Finally, marginal propensity to insure increases as insurers with lower net worth have higher operating leverage. Return on assets increase as capital prices drop and overall output decreases only slightly given the overall modest share of insurers in the entire equity market.

In the prior impulse response functions, the collateral constraint κ remained constant. As insurance coverage typically is a requirement prior to obtaining a mortgage, shocks to the insurers' net worth and credit constraints may co-move. Next to the disaster shock, I impose that the collateral constraint exogenously shifts from κ^H to κ^L . Figure 9 shows that exogenous shocks to collateral constraints amplify the effects of disasters on the economy. When the disaster shocks is paired with a collateral constraint shock, drops in consumption are steeper, savings increase as borrowing capacity tightens severely and both house and insurance prices drop sharply. These results reveal that the interaction between insurance market and borrowing conditions jointly shape how disasters transmit into the economy.

Impulse Response Functions

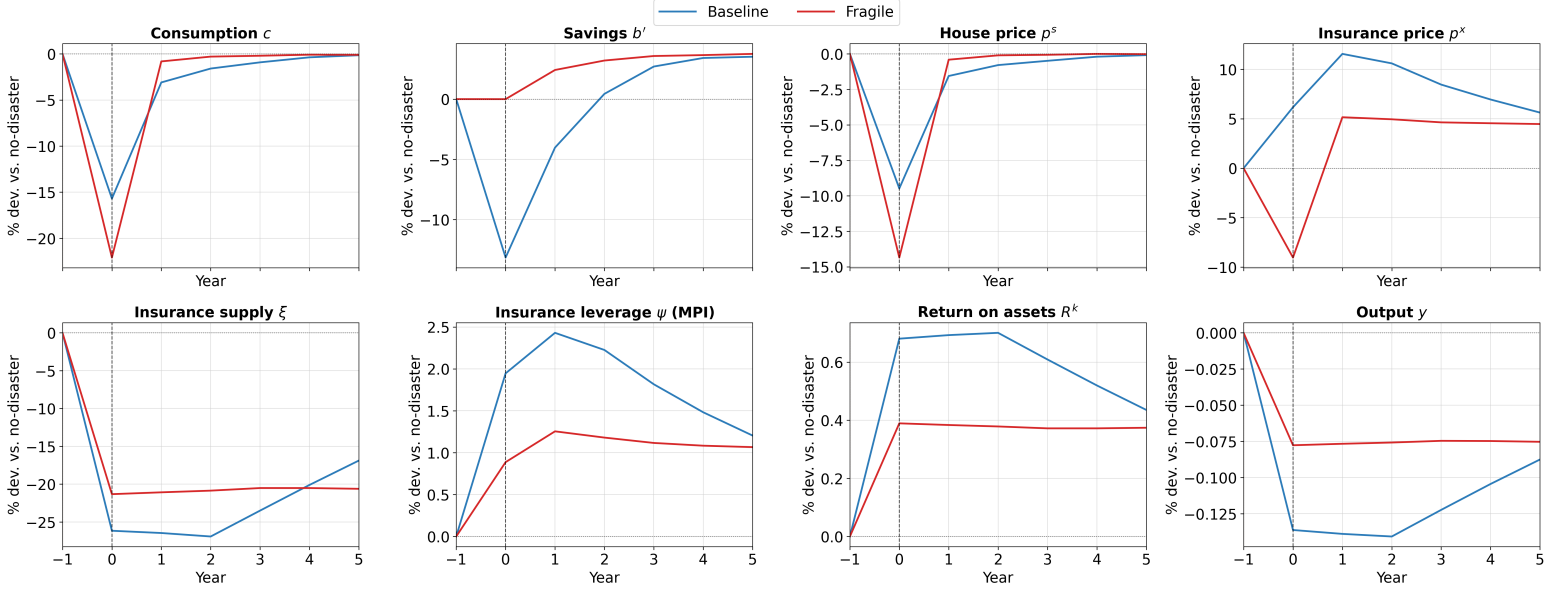


Figure 8. Impulse Response Functions to a Disaster Shock at T^* in the baseline and a fragile economy. In the fragile economy, insurer net worth is exogenously lowered prior to the arrival of the disaster shock at time $T^* - 1$. The straight and dashed lines correspond to the baseline and fragile economy, respectively.

Impulse Response Functions

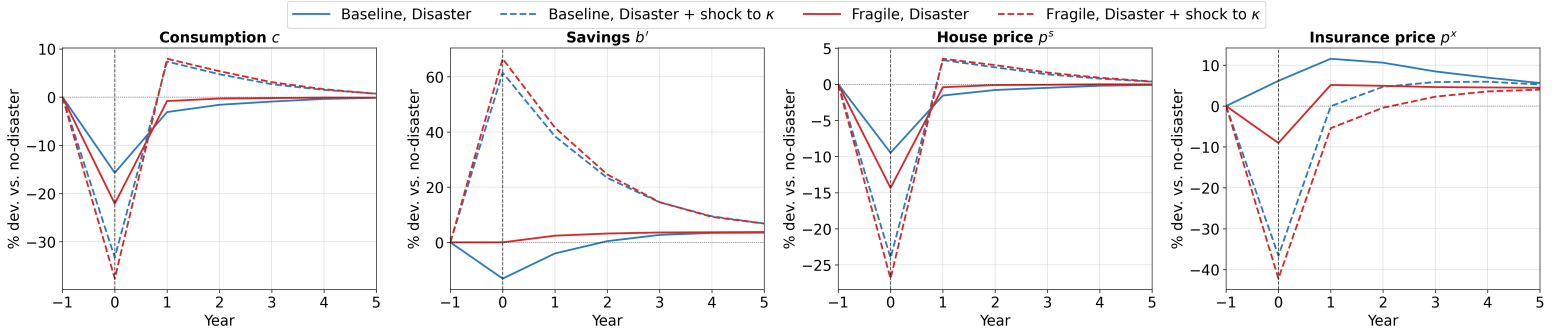


Figure 9. Impulse Response Functions to a Disaster Shock in the baseline and a fragile economy. In the fragile economy, insurer net worth is exogenously lowered prior to the arrival of the disaster shock.

7 Time-consistent insurance take-up

In the decentralized equilibrium, households take house prices as given. Consequently, they do not internalize how insurance decisions today affect house prices upon a disaster shock when the collateral constraint becomes binding, driving sharp reductions in house prices and consequently borrowing capacity. In addition, households take insurer net worth as given and do not internalize how high insurance demand today affects insurance prices and future insurance availability after disaster shocks when capital constraints become binding. The decentralized equilibrium therefore is characterized by two inefficiencies that a planner can

internalize and potentially improve long-run welfare. In the model, the planner aims to optimally the two occasionally binding constraints of the household and insurer.

In this section, I introduce a social planner who faces the same constraints as the household but internalizes the two externalities and derive the optimal and time-consistent insurance allocation. Section 7.1 presents the planner's problem and characterize the externalities. Section 7.2 outlines how the planner's solution can be decentralized using a Pigouvian tax on borrowing and insurance. Section 7.3 calibrates the social planner's problem and quantitatively compares the planner's solution to the decentralized equilibrium.

7.1 Planner's problem

I formulate the social planner's problem similar to the primal approach to optimal policy analysis. The planner chooses insurance x_{t+1} and b_{t+1} on behalf of the household subject to feasibility constraints, including the budget, collateral and insurer net worth constraint, and two implementability constraints, including the housing Euler equation (16). Since the planner cannot set prices directly, the planner is required to choose an allocation that satisfies the housing Euler equation. The Euler and insurance optimality equations of the DE do not enter the planner's problem as constraints because savings and insurance are controlled by the planner and their valuation may be different than the household's private valuation.

Households no longer decide on b_{t+1} and x_{t+1} and the household therefore takes as given the borrowing transfer $T_t^b = b_t - \frac{b_{t+1}}{R_t}$ and the net insurance transfer $T_t^x = \mathbf{1}_t \phi_t x_t p^r - p_t^x x_{t+1}$. Consequently, both the bond Euler equation and the insurance optimality condition drops from the optimization problem. As housing is in fixed supply, the social planner does not optimize over the housing choice. Household continue to choose home equity redraws optimally. While it would enter the planner's problem as a constraint, it never binds and can therefore be omitted (see proof).

The planner lacks the ability to commit to future policies. Hence, following [Bianchi and Mendoza \(2018\)](#) and [Biljanovska et al. \(2026\)](#), I focus on obtaining Markov stationary policy rules expressed as a function of the state variables $(b, x, n, \mathcal{S})$. The social planner takes as given the social planners' future choices. A time-consistent equilibrium is characterized by a fixed point at which the current decisions of the social planner coincide with the planner's future decisions.

Denote $\mathcal{X}(b, x, n, \mathcal{S})$ as the policy rules in the constrained-efficient equilibrium, and $\mathcal{C}(b, x, n, \mathcal{S})$, $\mathcal{B}(b, x, n, \mathcal{S})$, $l(b, x, n, \mathcal{S})$, $\mathcal{P}^s(b, x, n, \mathcal{S})$, $\mathcal{P}^x(b, x, n, \mathcal{S})$ and $\mu^*(b, x, n, \mathcal{S})$ the associated functions that return the values under that policy rule. The social planner's problem

can be represented in recursive form:

$$\begin{aligned}
\mathcal{V}(b, x, n, \mathcal{S}) &= \max_{c, l, x', b', p^s} u(c, s) + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \mathcal{V}(b', x', n', \mathcal{S}') \\
c + \frac{b'}{R} + p^x x' &= b + w + \mathbf{1}\phi [xp^r - 1] + l - f(l) + \Pi_t \\
-\frac{b'}{R} &\leq \kappa (p^s (1 - \mathbf{1}\phi) + \mathbf{1}\phi xp^r - l) \\
\zeta x' p^r &\leq V(n, \theta) \\
p^s u(c, 1) &= \beta \mathbb{E} \left[u'(c'(b', x', n', \mathcal{S}'), 1) \right. \\
&\quad \times \left\{ \nu_s g'(1) + p^{s'}(b', x', n', \mathcal{S}') \right\} \\
&\quad \left. + \left[p^{s'}(b', x', n', \mathcal{S}') (1 - \mathbf{1}'\phi') \right] \times \mu'(b', x', n', \mathcal{S}') \kappa' \right]
\end{aligned} \tag{37}$$

The planner economy's budget constraint has the multiplier $\lambda^* \geq 0$. The collateral constraint now has multiplier $\mu^* \geq 0$, which may be different from the private value of relaxing the constraint μ . The multiplier on the insurer leverage, private Euler, and implementability constraint are given by $\Psi^* \geq 0$, $\omega^* \geq 0$ and $\xi^* \geq 0$, respectively. While the planner takes as given the decisions of the future planners, it internalizes how current choices affect the planner's future decisions in the right hand side of implementability constraint, resulting in time consistent policy rules.

The insurance choice of the planner directly affect asset prices through current marginal utility. In addition, they affect asset prices indirectly by affecting the future planner's insurance coverage and insurer net worth, with their associated future allocations and prices.

The solution to the planner's problem is defined by the policy functions $b'(b, x, n, \mathcal{S})$, $x'(b, x, n, \mathcal{S})$ and their associated decision rules $c(b, x, n, \mathcal{S})$, $l(b, x, n, \mathcal{S})$, $\mu(b, x, n, \mathcal{S})$, pricing functions $p^s(b, x, n, \mathcal{S})$ and $p^x(b, x, n, \mathcal{S})$, and value function $V(b, x, n, \mathcal{S})$, the decision rules of future planners $\mathcal{B}(b, x, n, \mathcal{S})$ and $\mathcal{X}(b, x, n, \mathcal{S})$, and their associated decisions rules $\mathcal{C}(b, x, n, \mathcal{S})$, $\mathcal{L}(b, x, n, \mathcal{S})$, $\mu(b, x, n, \mathcal{S})$, and asset prices $\mathcal{P}^s(b, x, n, \mathcal{S})$ and $\mathcal{P}^x(b, x, n, \mathcal{S})$, such that the following conditions hold:

- Planner's optimization: the value function $V(b, x, n, \mathcal{S})$, and the decisions rules $b'(b, x, n, \mathcal{S})$, $x'(b, x, n, \mathcal{S})$, $c(b, x, n, \mathcal{S})$, $l(b, x, n, \mathcal{S})$, $\mu(b, x, n, \mathcal{S})$, $p^s(b, x, n, \mathcal{S})$, $p^x(b, x, n, \mathcal{S})$ solve the dynamic problem in (37) given the planner's future allocations $\mathcal{B}(b, x, n, \mathcal{S})$, $\mathcal{C}(b, x, n, \mathcal{S})$, $\mathcal{L}(b, x, n, \mathcal{S})$, $\mu(b, x, n, \mathcal{S})$, $\mathcal{P}^s(b, x, n, \mathcal{S})$, $\mathcal{P}^x(b, x, n, \mathcal{S})$.

- Time consistency: The current planner's optimal policy coincides with the conjectured policy rules governing the decisions of future planners, $b'(b, x, n, \mathcal{S}) = \mathcal{B}(b, x, n, \mathcal{S})$, $c(b, x, n, \mathcal{S}) = \mathcal{C}(b, x, n, \mathcal{S})$, $l(b, x, n, \mathcal{S}) = \mathbf{1}(b, x, n, \mathcal{S})$, $\mu(b, x, n, \mathcal{S}) = \mu(b, x, n, \mathcal{S})$, $p^s(b, x, n, \mathcal{S}) = \mathcal{P}^s(b, x, n, \mathcal{S})$, and $p^x(b, x, n, \mathcal{S}) = \mathcal{P}^x(b, x, n, \mathcal{S})$ for all $(b, x, n, \mathcal{S})$.

7.1.1 Comparison of equilibria

Taking the first-order conditions yields the following social planner's optimality conditions:

$$c_t :: \quad u'(t) = \lambda_t^* + u''(t)\xi_t^* p_t^s \quad (38)$$

$$b_{t+1} :: \quad \lambda_t^* = \beta R_t \mathbb{E}_{\mathcal{S}_{t+1}|\mathcal{S}_t} \left(\lambda_{t+1}^* + \xi_t^* \tilde{\Omega}_{t+1}^{b_{t+1}} \right) + \mu_t^* \quad (39)$$

$$x_{t+1} :: \quad p^x \lambda_t^* + \Psi_t^* \zeta p^r = \beta \mathbb{E}_{\mathcal{S}_{t+1}|\mathcal{S}_t} \left[\lambda_{t+1}^* + \mu_{t+1}^* \kappa_{t+1} \right] \mathbf{1}_{t+1} \phi_{t+1} - \Psi_{t+1}^* MPI_{t+1}^{con} \zeta p^r \Lambda_{x_{t+1}}^{n_{t+1}} + \xi_t^* \tilde{\Omega}_{t+1}^{x_{t+1}} \quad (40)$$

$$p^s :: \quad \xi_t^* = \frac{\kappa_t \mu_t^* (1 - \mathbf{1}_t \phi_t)}{u'(t)} \quad (41)$$

where

$$\Lambda_{x'}^{n'} = p^x (1 + R'^k) - \mathbf{1}' \phi' p^r - \zeta^1 \zeta^2 x'^{\zeta^2 - 1} \quad (42)$$

is the change in next-period insurer net worth given a small change in insurance take-up x_{t+1} . Compared to the first-order of consumption in the DE, the SP condition (38) has an additional term $u''(t)\xi_t^* p_t^s < 0$, which captures how an increase in consumption today affects house prices through the righthandside of the Housing . An increase in consumption directly raises asset prices and lowers the shadow cost of consumption.

Equation (39) have two additional terms worth mentioning. First, next to $\xi_t^* u''(t) p_t^s$ that is embedded in λ_t^* , the term $\lambda_{t+1}^* = u'(t+1) - \xi_{t+1}^* u''(t+1) p_{t+1}^s$ inside the expectation captures how borrowing today lowers consumption tomorrow and indirectly house prices by increasing the marginal utility of consumption at $t+1$. Second, the term $\Omega_{t+1}^{b_{t+1}}$ captures how borrowing today affects next-period house prices through the right-hand side of the implementability constraint.¹⁴

Equation (40) new relative to Bianchi and Mendoza (2018). It has a similar structure as the insurance optimality condition in the DE but differs in important ways. First, the term $-p^x \xi_t^* u''(t) p_t^s$ on the lefthandside (through λ_t^*) captures how insurance costs change

¹⁴ As shown in the Appendix, this terms consists of how borrowing today affects: i) future marginal utility and thus discounting of future house prices, 2) equilibrium house prices upon shocks through possible fire sales, and 3) collateral constraint tightness at time $t+1$.

marginal utility of consumption today and indirectly affect house prices.

Second, the marginal cost of insurance increases with Ψ_t^* . While the household in the DE is atomistic and does not consider how individual insurance demand affects capital constraints, the planner decides on aggregate insurance take-up. Doing so, the planner recognizes that high insurance take-up could make the insurer capital constrained, in which case the equilibrium insurance price contains the insurer's financial frictions. As Ψ^* increases the marginal cost of insurance when capital constraints bind, equilibrium insurance take-up is relatively low in the planner's economy when insurer net worth is low.

Third, the term $\xi_{t+1} u''(t+1) p_{t+1}^s \mathbf{1}_{t+1} \phi_{t+1}$ (through λ_{t+1}^*) captures how insurance today preserves consumption in the event of a disaster and indirectly affects tomorrow's house prices through tomorrow marginal utility. The term μ_{t+1}^* captures the social value of relaxing the expected future collateral constraints, which may be different from the private value μ_{t+1} .

Fourth, the term $MPI_{t+1}^{con} \zeta p^r \Lambda_{x'}^{n'}$ captures how insurance take-up today leads to possible future insurers' binding capital constraints, evaluated at shadow value Ψ_{t+1}^* . When the planner increases insurance take-up today, insurer operating leverage increases and the net worth reduction in disaster states becomes larger. When the disaster states realizes, the planner suffers from high insurance prices and low equilibrium insurance quantities. The correction term is proportional to MPI_{t+1}^{con} , as this captures how future insurance supply changes with changes in future net worth. Furthermore, the correction term is proportional to ζp^r as higher net worth precisely relaxes future insurance supply by ζp^r (see equation (24)). Finally, it is proportional to $\Lambda_{x'}^{n'}$, which captures how insurers' future net worth changes given a small increase in insurance take-up today. This correction term could be either positive or negative when more insurance respectively eases or tightens future capital constraints in expectation.

Finally, $\Omega_{t+1}^{x_{t+1}}$ captures how insurance affects house prices tomorrow through the righthand-side of the implementability constraint. Focusing on $\Omega_{t+1}^{x_{t+1}}$, the first term reflects how insurance affect future marginal utility and therefore future asset prices. Since higher x' raises c' in the event of a disaster, and given $u''(c) < 0$, this term is positive. The second term captures how insurance holdings affect asset prices and insurance payouts. Insurance take-up increase collateral values in the event of a disaster and reduces fire sales. The third term captures how insurance holdings affect the tightness of the collateral constraint at time $t+1$, which in turns affects house prices at time t . This term is negative. Fourthly, the planner considers how insurance take-up affects house price through the future net worth of the insurer. Quantitatively, the model shows that house prices are higher when insurers

have high net worth, which is partly driven by lower insurance prices in these states.

Overall, the marginal benefit of insurance according to the social planner thus depends on how insurance protects wealth and prevents fire sales in the event of a disaster corrected for how insurance today affects future capital constraints and future insurance availability through insurers' marginal propensity to insure.

7.2 Pigouvian tax on insurance

Appendix 10.4 shows that the social planner's allocation can be decentralized with a tax on insurance, which equals the ratio of the insurance price in the DE to that of in the planner's economy minus 1:

$$\tau^x = \frac{(p_t^x)^{DE}}{(p_t^x)^{PL}} - 1 \quad (43)$$

When the planner values insurance more than in the decentralized economy, leading to relatively high insurance prices, the tax becomes negative. Note that both the numerator and denominator are evaluated at the planner's policy functions as when the tax is implemented the planner's and decentralized equilibrium's insurance choice coincide. Whenever the implementability and feasibility constraints today do not bind, one can derive the macro-prudential (MP) insurance tax:

$$\tau_{MP}^x = \frac{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} [(u'(c', 1) + \mu' \kappa') \mathbf{1}' \phi' p^r]}{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} [(\lambda'^* + \mu'^* \kappa') \mathbf{1}' \phi' p^r + \Psi'^* MPI'^{con} \zeta p^r \Lambda_{x'}^{n'}]} - 1 \quad (44)$$

The macro-prudential (MP) insurance tax is levied in good times when household collateral constraints and insurer capital constraints to now bind, but can bind with positive probability at time $t+1$. Furthermore, an insurance tax can be levied even when there is zero probability of hitting future constraints when the planner's and the DE economy have different levels of consumption, which affect marginal utilities across the state space and thus the valuation of insurance.

The macro-prudential tax consists of two parts. First, the tax depends on how the planner and the unconstrained household differently value the benefit of insurance upon a disaster. Both the numerator and the denominator share the term $\mathbf{1}' \phi' p^r$, which captures the insurance payout at a disaster with full insurance. Importantly, the unconstrained household and planner value this possible future insurance payout differently. Whenever $\lambda'^* + \mu'^* \kappa' > u'(c', 1) + \mu' \kappa'$, the marginal benefit of insurance in the planner economy exceeds that of the unconstrained household, calling for a subsidy.

Second, the tax depends on how insurance take-up today affects future capital constraints evaluated at shadow cost Ψ_{t+1}^* . When the term $\mathbb{E}_{S'|S} [\Psi'^* MPI'^{con} \zeta p^r \Lambda_{x'}^{n'}]$ is negative, more insurance today is costly in expectation given future possible capital constraints. The correction term then lowers insurance take-up by increasing the tax.

The macro-prudential insurance tax thus broadly consists of two components. One component internalizes how insurance affects house prices upon a disaster shock by mitigating fire sales. The second component internalizes how more insurance today leads to possible binding insurance availability constraint in the next period. Overall the sign on the tax is ambiguous. If the former effect dominates, it calls for a insurance subsidy, whereas if the second effect dominates, it calls for a tax.

Similarly, one can derive the optimal tax on borrowing, which is similar to the tax derived in [Bianchi and Mendoza \(2018\)](#) and [Biljanovska et al. \(2026\)](#):

$$1 + \tau^b = \frac{\beta R \mathbb{E}_{S'|S} (u'(c', 1) - \xi'^* u''(c', 1) p'^s + \xi^* \hat{\Omega}_{b'}') + \xi^* u''(c, 1) p^s}{\beta R \mathbb{E}_{S'|S} u'(c', 1)} \quad (45)$$

7.3 Quantitative analysis of the social planner

Policy functions Figure 10 compares the policy functions of the planner with that of in the decentralized equilibrium. There are various notable differences. The insurance choice absent a disaster of the planner is higher than that of the unconstrained household, and the difference becomes larger for low levels of savings (high levels of debt). When more indebted, the possibility of experiencing future binding collateral constraints is high, and the value of insurance increases as the planner internalizes possible house price crashes that lead to severe consumption drops. Furthermore, savings in the no disaster state are higher in the planner economy. While the differences are small, the non-linear interaction between debt and house prices upon a disaster make that these changes can have a meaningful effect on how severely consumption drops upon a disaster.

Taxation Figure 11 plots the macro-prudential insurance and debt tax across the savings state, and separately for different levels of net worth. Consistent with [Bianchi and Mendoza \(2018\)](#), the debt tax is exclusively positive and decreases in savings. The level of debt taxes is lower than in [Bianchi and Mendoza \(2018\)](#), who find an optimal debt tax of up to 8%. As disaster risk can be mitigated by both lower debt and more insurance, when both corrections are at play, the magnitude of the debt correction is smaller.

Figure 12 decomposes the tax into the effects stemming from different valuations of the

Planner policy functions

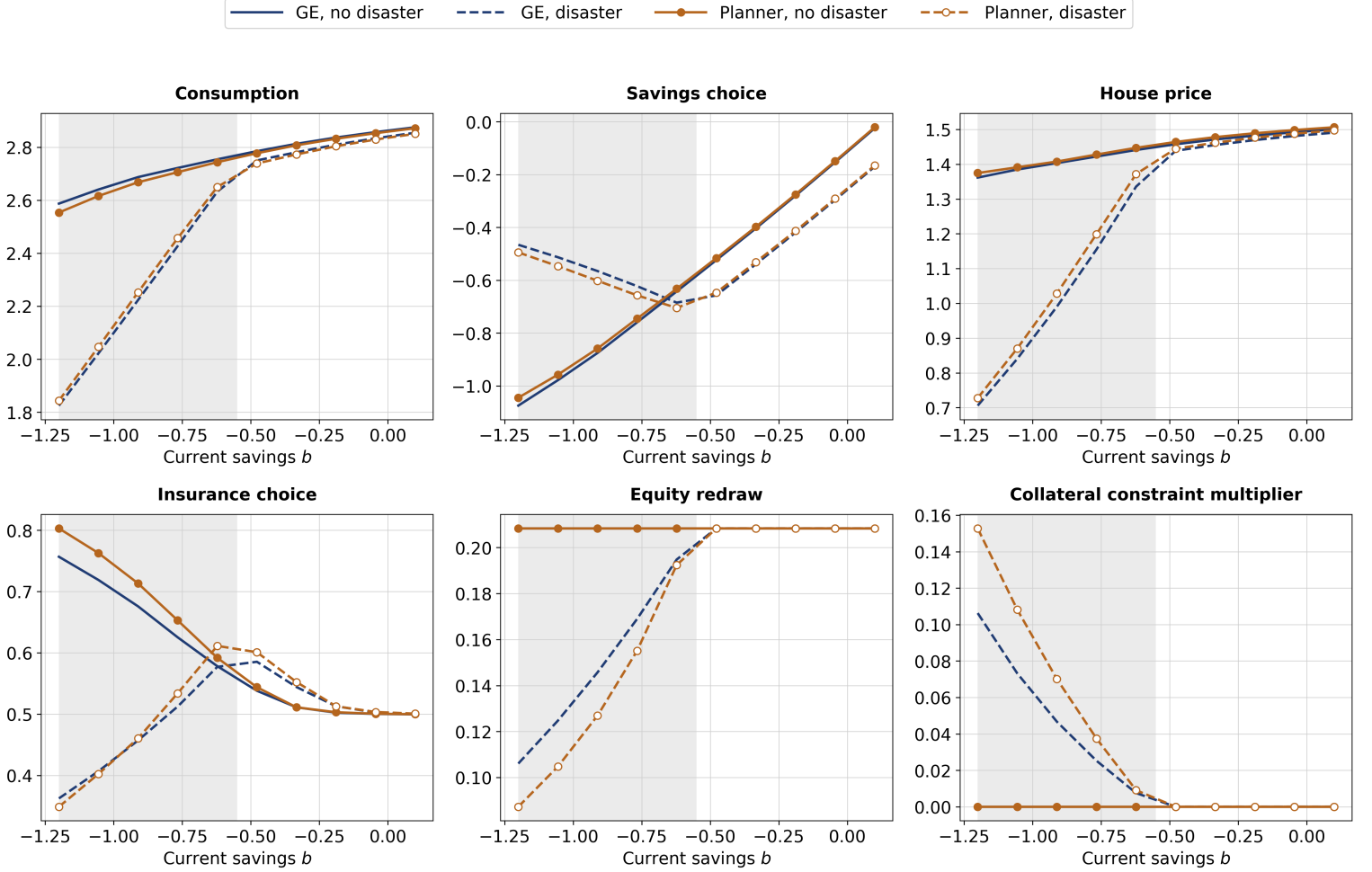


Figure 10. Comparison of policy functions in the decentralized and constrained-efficient equilibrium.

benefit of insurance and the channel that internalizes how insurance today affect future possible capital constraints. At low levels of savings, the subsidy is highest and almost exclusively driven by differences in the valuation of insurance. At medium levels of savings, the optimal subsidy is lowered driven high insurance take-up causing a higher likelihood of experiencing binding capital constraints in the future. At high levels of savings, only the insurer net worth wedge is at play. The magnitude of it is however small since insurance take-up in high savings state is small to begin with (see Figure 10).

Table 5 presents the results of the simulations in both the planner and decentralized equilibrium. Welfare improvements are small through not negligible, amounting to 0.03% of consumption equivalents. The debt tax averages 0.43%, resulting in smaller levels of debt taken on relative to home valuations, and the house price decline upon a disaster in the planner's economy is significantly smaller.

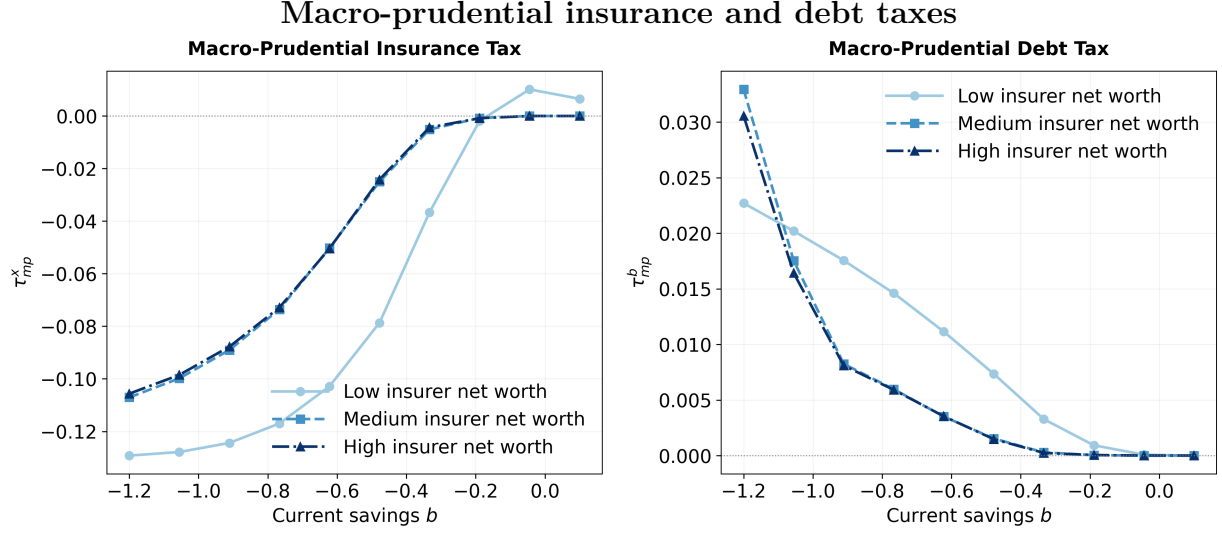


Figure 11. Macro-prudential insurance and debt taxes across the savings state for different levels of net worth. The insurance state is set to the median, and I consider the no-disaster and high collateral constraint state.

Decomposition of Macro-prudential Insurance Taxes across Savings Levels

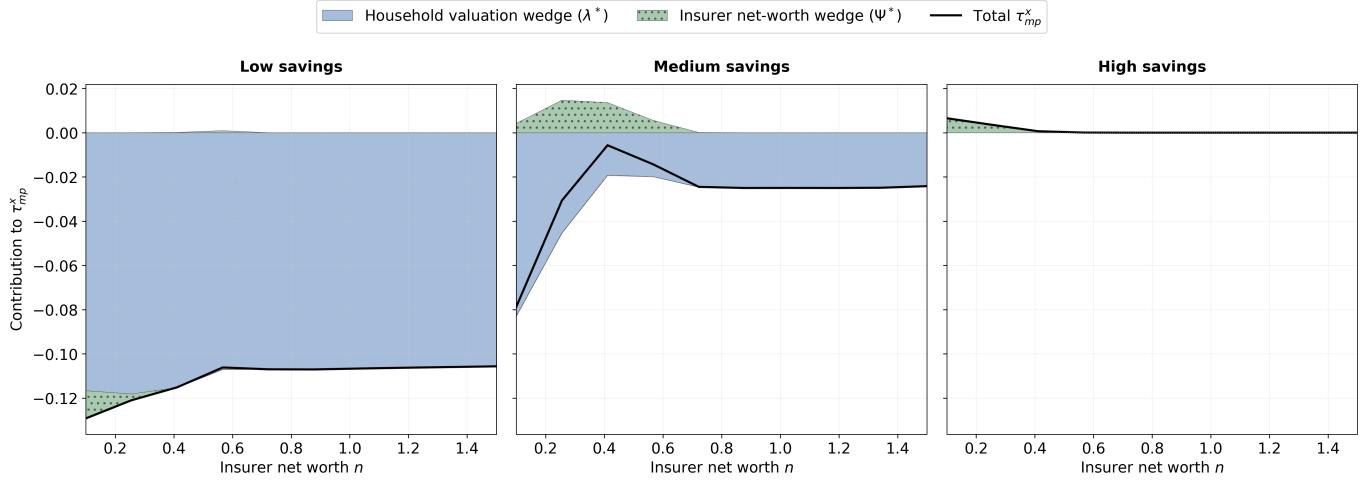


Figure 12. Decomposition of the Macro-prudential insurance tax into the household valuation wedge and the insurer net worth wedge. The household valuation wedge captures differences in future marginal utilities when a disaster hits. The insurer net worth wedge corrects for the possibility of future binding capital constraints.

8 Conclusion

The availability of homeowners insurance in weather exposed housing markets has become uncertain, posing risks for homeowners and broader financial stability. Existing insurance regulations tend to focus on preserving household wealth upon weather shocks, but ignore how insurance take-up today affect the long-run availability of insurance. This paper makes

Table 5
Comparison of equilibria and welfare gains

	Decentralized Equilibrium		Social Planner	
	Mean	Std. Dev.	Mean	Std. Dev.
<i>Summary statistics</i>				
House prices	1.419	0.053	1.432	0.040
Consumption	2.721	0.066	2.712	0.055
Output	4.645	0.011	4.640	0.011
Marginal Propensity to Insure	0.495	0.626	0.795	0.605
Insurance premium p^x	0.027	0.001	0.023	0.003
Insurance gap	49.3%	—	60.3%	—
New debt to house price	45.5%	—	37.5%	—
Share of periods collateral constraint binds	5.3%	—	5.3%	—
Share of periods insurer constraint binds	37.2%	—	62.5%	—
House price decline upon disaster	16.2%	—	10.9%	—
Insurer net worth decline upon disaster	25.9%	—	28.1%	—
<i>Mean tax and welfare gains</i>				
Debt tax τ^b	—	—	0.43%	0.14%
Insurance tax τ^x	—	—	−4.84%	1.55%
ρ_{τ^b, τ^x}	—	—	−0.89	—
Welfare gains	—	—	0.026%	—

Decomposition of Total Insurance Taxes across Savings Levels

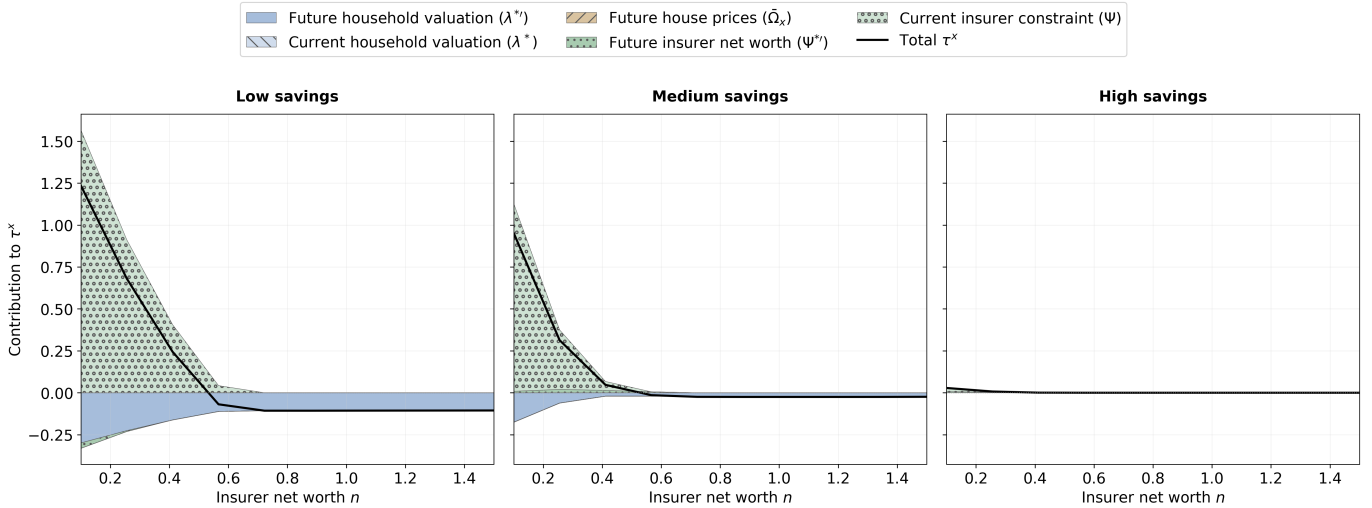


Figure 13. Decomposition of total insurance taxes.

a first attempt to derive the optimal and time-consistent insurance subsidy or tax that optimizes long-run household welfare while accounting for insurer financial constraints.

References

- Abramson, B., P. Sastry, and I. Sen**, “Residual Markets,” 2025. Working paper.
- Aon**, “2026 Climate and Catastrophe Insight,” 2026.
- Becker, Bo and Marcus Opp**, “Regulatory reform and risk-taking: Replacing ratings,” Technical Report, National Bureau of Economic Research 2013.
- Bianchi, Javier and Enrique G Mendoza**, “Optimal time-consistent macroprudential policy,” *Journal of Political Economy*, 2018, 126 (2), 588–634.
- Biljanovska, Nina, Jordi Galí, Lucyna Górnicka, and Alexandros P Vardoulakis**, “A Model of Leveraged Bubbles,” Technical Report, National Bureau of Economic Research 2026.
- Boomhower, Judson, Meredith Fowlie, Jacob Gellman, and Andrew Plantinga**, “How Are Insurance Markets Adapting to Climate Change? Risk Classification and Pricing in the Market for Homeowners Insurance,” Technical Report, National Bureau of Economic Research 2024.
- Born, Patricia H and Barbara Klimaszewski-Blettner**, “Should I stay or should I go? The impact of natural disasters and regulation on US property insurers’ supply decisions,” *Journal of Risk and Insurance*, 2013, 80 (1), 1–36.
- Coimbra, Nuno and H  lene Rey**, “Financial cycles with heterogeneous intermediaries,” *Review of Economic Studies*, 2024, 91 (2), 817–857.
- Cookson, J Anthony, Emily Gallagher, and Philip Mulder**, “Coverage neglect in homeowners insurance,” *Available at SSRN 5057551*, 2024.
- Costa, Rodrigo and Jack W Baker**, “A methodology to estimate postdisaster unmet housing needs using limited data: Application to the 2017 California wildfires,” *Risk Analysis*, 2024, 44 (4), 850–867.
- Eastman, Evan, Kyeonghee Kim, and Tingyu Zhou**, “Homeowners insurance and housing prices,” *Available at SSRN*, 2024.
- ECB-EIOPA**, “Policy options to reduce the climate insurance protection gap,” 2023.

- Favara, Giovanni and Jean Imbs**, “Credit supply and the price of housing,” *American Economic Review*, 2015, 105 (3), 958–992.
- Freddie Mac**, “Trends in Mortgage Refinancing Activity,” <https://www.freddiemac.com/research/insight/20220425-trends-mortgage-refinancing-activity> April 2022. Freddie Mac Research.
- Gallagher, Justin**, “Learning about an infrequent event: Evidence from flood insurance take-up in the United States,” *American Economic Journal: Applied Economics*, 2014, pp. 206–233.
- Ge, Shan and Michael S Weisbach**, “The role of financial conditions in portfolio choices: The case of insurers,” *Journal of Financial Economics*, 2021, 142 (2), 803–830.
- Gertler, Mark and Nobuhiro Kiyotaki**, “Financial intermediation and credit policy in business cycle analysis,” in “Handbook of monetary economics,” Vol. 3, Elsevier, 2010, pp. 547–599.
- Iacoviello, Matteo**, “House prices, borrowing constraints, and monetary policy in the business cycle,” *American economic review*, 2005, 95 (3), 739–764.
- Ibragimov, Rustam, Dwight Jaffee, and Johan Walden**, “Nondiversification traps in catastrophe insurance markets,” *The Review of Financial Studies*, 2009, 22 (3), 959–993.
- Jamilov, Rustam and Tommaso Monacelli**, “Bewley banks,” *Review of Economic Studies*, 2025, p. rdaf062.
- Koijen, Ralph SJ and Motohiro Yogo**, “The cost of financial frictions for life insurers,” *American Economic Review*, 2015, 105 (1), 445–475.
- and —, “The fragility of market risk insurance,” *The Journal of Finance*, 2022, 77 (2), 815–862.
- Kunreuther, Howard and Mark Pauly**, “Neglecting disaster: Why don’t people insure against large losses?,” *Journal of Risk and Uncertainty*, 2004, 28 (1), 5–21.
- Liu, Jessica and Weiling Liu**, “The effect of political frictions on the pricing and supply of insurance,” *The Review of Financial Studies*, 2024, 37 (4), 1149–1189.

- Mendoza, Enrique G and Sergio Villalvazo**, “FiPIt: A simple, fast global method for solving models with two endogenous states & occasionally binding constraints,” *Review of Economic Dynamics*, 2020, 37, 81–102.
- Oh, Sangmin, Ishita Sen, and Ana-Maria Tenekedjieva**, “Pricing of climate risk insurance: Regulation and cross-subsidies,” *Journal of Finance*, *Forthcoming*, 2021.
- Sastry, Parinitha, Ishita Sen, Ana-Maria Tenekedjieva, and Therese C Scharlemann**, “The limits of insurance demand and the growing protection gap,” Technical Report, Working Paper 2025.
- , —, —, and —, “The Economics of Insurance Guaranty Funds,” *Available at SSRN 5806662*, 2025.
- Solomon, Adam**, “Insuring Correlated Climate Risk: Evidence from Public Reinsurance,” *Available at SSRN 4881751*, 2024.
- U.S. Bureau of Labor Statistics**, “Consumer Expenditures – 2024,” News Release USDL-25-1586, U.S. Department of Labor December 2025.
- Zivin, Joshua Graff, Yanjun Liao, and Yann Panassie**, “How hurricanes sweep up housing markets: Evidence from Florida,” *Journal of Environmental Economics and Management*, 2023, 118, 102770.

9 Empirics

Effects of insurer non-renewals on house prices by instrument distance cutoff

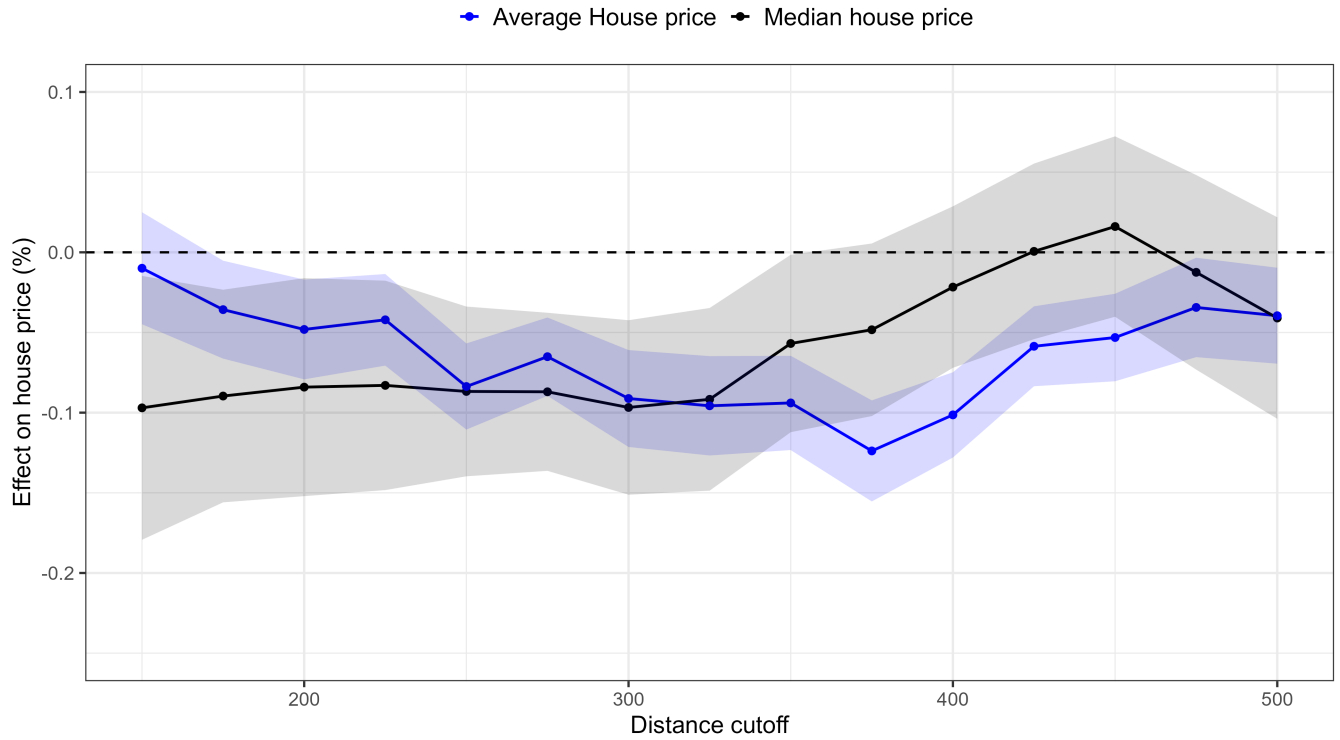


Figure B.1. Second stage coefficient estimates in a regression of log house prices (median or average) on the log of insurer-driven non-renewals at the zip-code level, which is instrumented with out-of-zip-code weather damages weighted by the incumbent insurers' out-of-zip-code portfolio shares. The regression is ran for different distance cutoffs, that determine which damages are included in the instrument. Standard errors are clustered at the zip-code level.

Kernel Density plot of auto-correlation and volatility in disaster damages

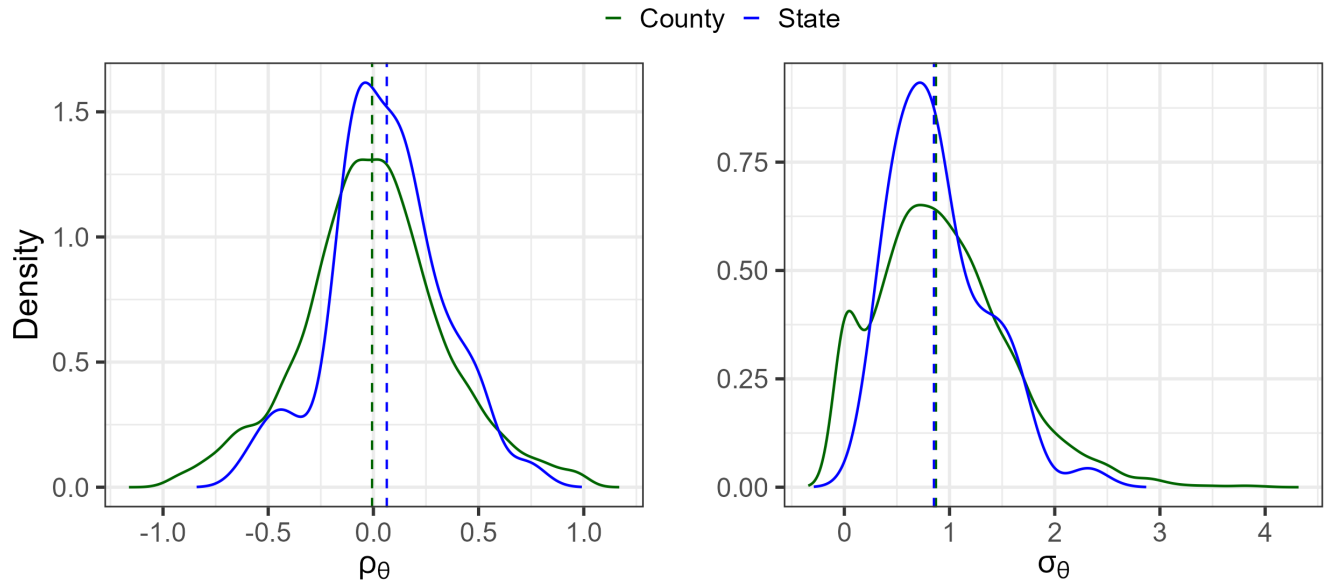


Figure B.2. Autocorrelation (left) and volatility (right) is computed at both the state and county level in the period 1994 - 2024. Dashed vertical line shows median autocorrelation and volatility coefficient by state and county. Disaster severity is calculated as dollar damages divided by total income at the county or state level. All types of disasters are included.

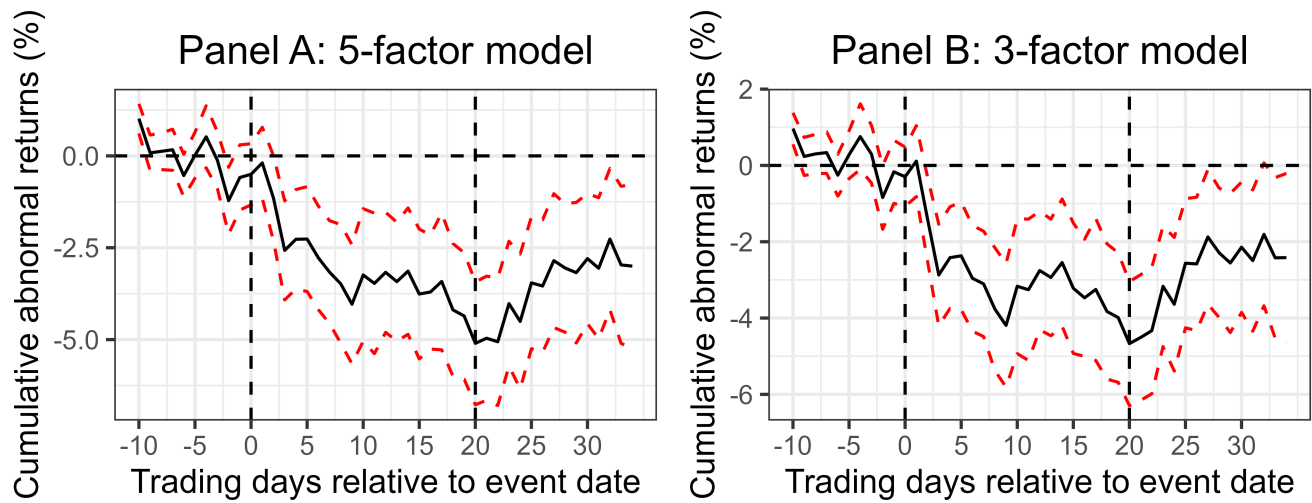


Figure B.3. Insurer stock returns after the removal of NOAA climate data. 20 days after the announcement, the data became available to insurers again.

Co-movement in insurance prices (Florida) and reinsurance prices

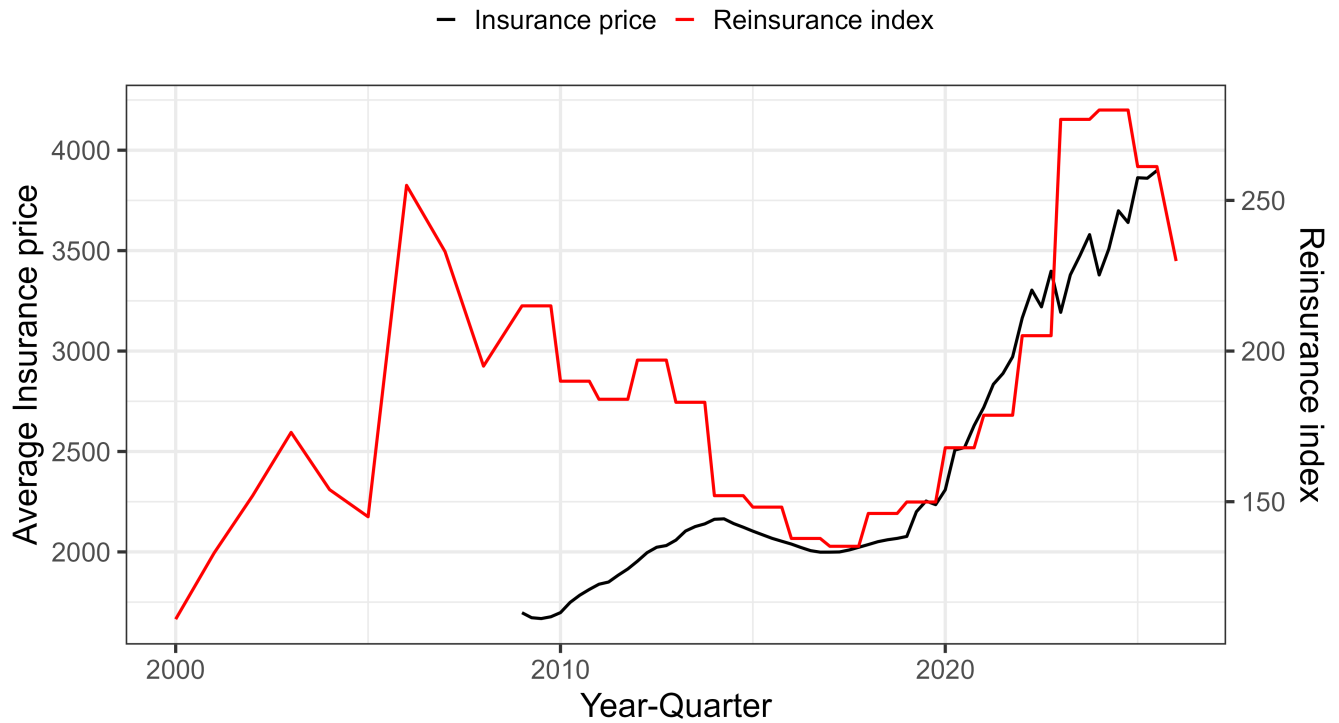


Table A.1

Empirical estimates of marginal propensity to insure (panel A) and exit (panel B). Standard errors in parentheses are clustered at the insurer level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

Panel A				
Model:	Log Exposure		Δ Log Exposure	
	(1)	(2)	(3)	(4)
Log Net Worth	0.27*** (0.10)	0.75*** (0.15)		
Δ Log Net Worth			0.62*** (0.17)	0.08 (0.16)
Zip-code $\times$ Year FE	Yes	Yes	Yes	Yes
Insurer FE	No	Yes	No	Yes
Observations	724,911	724,911	625,809	625,809
R ²	0.24	0.61	0.06	0.24
Within R ²	0.10	0.03	0.03	0.0004
Panel B				
Model:	(1)	(2)	Exit _{t+1} (3)	(4)
Log Net Worth	-0.009** (0.004)	-0.09*** (0.02)		
Δ Log Net Worth			-0.10** (0.05)	0.03 (0.05)
Zip-code $\times$ Year FE	Yes	Yes	Yes	Yes
Insurer FE	No	Yes	No	Yes
Observations	665,018	665,018	665,018	665,018
R ²	0.09	0.21	0.09	0.20
Within R ²	0.008	0.02	0.003	0.0002

Table A.2

Heterogeneity in marginal propensity to insure. Standard errors in parentheses are clustered at the insurer level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

	Δ Log Exposure			
	(1)	(2)	(3)	(4)
Δ Log Net Worth	1.9** (0.79)	-0.74 (0.55)	0.69*** (0.18)	1.3 (0.91)
Δ Log Net Worth $\times$ Log RBC Ratio	-0.19* (0.12)			
Δ Log Net Worth $\times$ Leverage		2.2** (0.87)		
Δ Log Net Worth $\times$ Market Share			-0.03*** (0.008)	
Δ Log Net Worth $\times$ Log Total Assets				-0.05 (0.06)
Zip-code $\times$ Year FE	Yes	Yes	Yes	Yes
Insurer FE	No	No	No	No
Observations	560,156	625,809	625,809	625,809
R ²	0.07	0.07	0.06	0.06
Within R ²	0.04	0.04	0.03	0.03

Table A.3

The role of insurer financial conditions in insurance prices and non-renewals. Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

	Δ Log Ins. Price				Log Non-renewals			
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)	OLS (7)	IV (8)
Δ Log Net Worth (avg)	-0.19** (0.09)	-2.9** (1.2)			-0.35** (0.16)	-3.8* (2.1)		
Δ Log RBC (avg)			0.004 (0.03)	-0.45*** (0.17)			-0.35*** (0.07)	-0.57** (0.28)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip-Code FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,046	15,046	14,935	14,935	8,043	8,029	8,030	8,018
R ²	0.16	0.16	0.18	0.18	0.96	0.96	0.96	0.96
Within R ²	0.01	0.01	0.01	0.01	0.04	0.05	0.05	0.05

Table A.4

OLS Regression of house prices on the share of the residual insurer in California (FAIR Plan). Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

	Log(Median HP) (1)	Log(Mean HP) (2)
FAIR Plan Share	-0.41*** (0.07)	-0.25*** (0.04)
Controls	Yes	Yes
Zip-code FE	Yes	Yes
Year FE	Yes	Yes
Observations	12,768	11,753
R ²	0.96	0.99
Within R ²	0.09	0.16

10 Model

10.1 Household optimality conditions - decentralized equilibrium

The Lagrangian for the household is:

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k \left\{ \frac{1}{1-\psi} (c_{t+k} + \nu_s g(s_{t+k}))^{1-\psi} \right. \\ & + \lambda_{t+k} \left[b_{t+k} + h_{t+k} w_{t+k} + p_{t+k}^s s_{t+k} + \mathbf{1}_{t+k} \phi_{t+k} s_{t+k} (x_{t+k} - 1) + l_{t+k} - f(l_{t+k}) \right. \\ & \quad \left. \left. - c_{t+k} - \frac{b_{t+1+k}}{R_{t+k}} - s_{t+1+k} p_{t+k}^s - p_{t+k}^x x_{t+1+k} s_{t+1+k} p^r \right] \right. \\ & \left. + \mu_{t+k} \left[\kappa_{t+k}(x_{t+1+k}) \times \left(p_{t+k}^s s_{t+k} (1 - \mathbf{1}_{t+k} \phi_{t+k}) + \mathbf{1}_{t+k} \phi_{t+k} s_{t+k} x_{t+k} p^r - l_{t+k} \right) + \frac{b_{t+1+k}}{R_{t+k}} \right] \right\} \end{aligned}$$

Define the composite $\mathcal{X}_t = c_t + \nu_s g(s_t)$ and noting that $\lambda_t = \mathcal{X}_t^{-\psi} = U'(t)$, the optimality condition for borrowing is:

$$U'(t) = \beta R_t \mathbb{E}_t [U'(t+1)] + \mu_t \quad (46)$$

The first-order condition for insurance demand x_{t+1} yields the following equation:

$$-\lambda_t p_t^x s_{t+1} + \beta \mathbb{E}_t [\mathbf{1}_{t+1} \phi_{t+1} p^r (\lambda_{t+1} + \mu_{t+1})] s_{t+1} = 0 \quad (47)$$

Rearranging gives:

$$p_t^x \lambda_t = \beta \mathbb{E}_t [\mathbf{1}_{t+1} \phi_{t+1} p^r (\lambda_{t+1} + \mu_{t+1} \kappa_{t+1})] \quad (48)$$

The first-order condition for housing s_{t+1} yields:

$$\begin{aligned} \lambda_t (p_t^s + p_t^x x_{t+1}) = & \beta \mathbb{E}_t \left[\lambda_{t+1} \left\{ \nu_s g'(s_{t+1}) + p_{t+1}^s + \mathbf{1}_{t+1} \phi_{t+1} x_{t+1} p^r \right\} \right. \\ & \left. + \mu_{t+1} \kappa_{t+1} (p_{t+1}^s (1 - \mathbf{1}_{t+1} \phi_{t+1}) + \mathbf{1}_{t+1} \phi_{t+1} x_{t+1} p^r) \right] \end{aligned} \quad (49)$$

Substituting the insurance condition into the housing condition gives:

$$\lambda_t p_t^s = \beta \mathbb{E}_t \left[\lambda_{t+1} \left\{ \nu_s g'(s_{t+1}) + p_{t+1}^s \right\} + \mu_{t+1} \kappa_{t+1} (p_{t+1}^s (1 - \mathbf{1}_{t+1} \phi_{t+1})) \right] \quad (50)$$

The first order conditions for home equity redraws l_t gives:

$$1 = \theta l_t + \frac{\mu_t \kappa_t}{\lambda_t} \quad (51)$$

Derivation house price If the collateral constraint does not bind, i.e., $\mu_t = 0$, then equation (16) yields:

$$q_t^s = \mathbb{E}_t \left[m_{t,t+1} \left(\nu_s g'(s_{t+1}) + q_{t+1}^s \right) \right] \quad (52)$$

Iterating one-period forward gives:

$$q_{t+1}^s = \mathbb{E}_{t+1} \left[m_{t+1,t+2} \left(\nu_s g'(s_{t+2}) + q_{t+2}^s \right) \right] \quad (53)$$

Substituting back and using the law of iterated expectations yields:

$$q_t^s = \mathbb{E}_t \left[m_{t,t+1} \nu_s g'(s_{t+1}) \right] + \mathbb{E}_t \left[m_{t,t+1} m_{t+1,t+2} \nu_s g'(s_{t+2}) \right] + \mathbb{E}_t \left[m_{t,t+1} m_{t+1,t+2} q_{t+2}^s \right] \quad (54)$$

Iterating this forward and using the transversality condition, one can express the value of the housing investment today as the expected present value of housing gains (marginal housing utility) discounted with R_{t+1}^s :

$$q_t^s = \mathbb{E}_t \left[\sum_{j=0}^{\infty} \left(\prod_{i=0}^j \mathbb{E}_{t+i} R_{t+1+i}^s \right)^{-1} \nu_s g'(s_{t+j+1}) \right] \quad (55)$$

Excess housing return We can combine equations (13) and (16) to derive the expected excess return on housing $R^{hp} \equiv \mathbb{E}_t [R_{t+1}^s - R_t]$. Note that (13) can be written as:

$$1 = \mathbb{E}_t [m_{t+1}] R_t + \frac{\mu_t}{\lambda_t} \quad (56)$$

Similarly, (16) can be written as:

$$1 = \mathbb{E}_t \left[m_{t+1} \frac{\nu_s g'(s_{t+1}) + q_{t+1}^s}{q_t^s} \right] + \beta \mathbb{E}_t \left[\frac{\mu_{t+1} \kappa_{t+1} q_{t+1}^s}{\lambda_t q_t^s} \right] \quad (57)$$

$$1 = \mathbb{E}_t [m_{t+1} R_{t+1}^s] + \mathbb{E}_t [m_{t+1} \Upsilon_{t+1}] \quad (58)$$

Subtracting both equations yields:

$$\mathbb{E}_t \left[m_{t+1} (R_{t+1}^s - R_t) \right] = \frac{\mu_t}{\lambda_t} - \mathbb{E}_t [m_{t+1} \Upsilon_{t+1}] \quad (59)$$

$$R_t^{hp} = \mathbb{E}_t [R_{t+1}^s - R_t] = \underbrace{\frac{\mu_t}{\lambda_t \mathbb{E}_t [m_{t+1}]}_{\text{Liquidity premium}}} - \underbrace{\frac{\mathbb{E}_t [m_{t+1} \Upsilon_{t+1}]}{\mathbb{E}_t [m_{t+1}]}_{\text{Collateral effect}}} - \underbrace{\frac{\text{Cov}_t (m_{t+1}, R_{t+1}^s)}{\mathbb{E}_t [m_{t+1}]}_{\text{Risk premium}}} \quad (60)$$

where

$$\Upsilon_{t+1} = \frac{\mu_{t+1} \kappa_{t+1} q_{t+1}^s}{\lambda_{t+1} q_t^s} \quad (61)$$

10.2 Goods market clearing

Start from the household budget constraint:

$$c = b - b'/R - p^x x' + hw - p^s \mathbf{1}\phi + \mathbf{1}\phi x + \Pi \quad (62)$$

Substitute the final good firm's sectoral constraint $y = wh + q^a R^k \Phi(a)$ gives:

$$c = y + b - b'/R - p^x x' - q^a R^k \Phi(a) - p^s \mathbf{1}\phi + \mathbf{1}\phi x + \Pi \quad (63)$$

Substituting $\Pi = n_t - N_t$ and noting that $n_t = a_t(1 + R^k) - \mathbf{1}\phi x$ gives:

$$c = y + b - b'/R - p^x x' - q^a R^k \Phi(a) - p^s \mathbf{1}\phi + a_t(1 + R^k) - N_t \quad (64)$$

Substituting the insurer balance sheet constraint $N_t = a - p^x x'$ gives:

$$c = y + b - b'/R - p^s \mathbf{1}\phi + R^k(a_t - q^a \Phi(a)) \quad (65)$$

Note that the perfect competition argument results in the last term dropping out:

$$c = y + b - b'/R - p^s \mathbf{1}\phi \quad (66)$$

10.3 Planner optimality conditions

The planner's problem in recursive form can be formulated as below.

$$\begin{aligned}
\mathcal{L} = & u(c, 1) + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} [V(b', x', n', \mathcal{S}')] \\
& + \lambda^* \left[b + w + \mathbf{1}\phi [xp^r - 1] + l - f(l) + \Pi - c - \frac{b'}{R} - p^x x' \right] \\
& + \mu^* \left[\kappa \left(p^s (1 - \mathbf{1}\phi) + \mathbf{1}\phi xp^r - l \right) + \frac{b'}{R} \right] \\
& + \xi^* \left[\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[u'(c'(b', x', n', \mathcal{S}'), 1) \left\{ \nu_s g'(1) + Q'(b', x', n', \mathcal{S}') \right\} \right. \right. \\
& \left. \left. + \mu'(b', x', n', \mathcal{S}') \kappa' Q'(b', x', n', \mathcal{S}') \right] - u'(c, 1) p^s \right] \\
& + \Psi^* \left[V(n, \theta) - x' \zeta p^r \right]
\end{aligned} \tag{67}$$

Taking the first-order condition with respect to c yields:

$$u'(c, 1) = \lambda^* + u''(c, 1) \xi^* p^s \tag{68}$$

The first-order conditions for b' yields:

$$\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \lambda^{*'} - \frac{\lambda^*}{R} + \frac{\mu^*}{R} + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} [\xi^* \tilde{\Omega}'_{b'}] = 0 \tag{69}$$

where

$$\begin{aligned}
\tilde{\Omega}'_{b'} = & u''(c'(b', x', n', \mathcal{S}'), 1) (\nu_s g'(1) + Q'(b', x', n', \mathcal{S}') + \mu'(b', x', n', \mathcal{S}') \kappa' Q'(b', x', n', \mathcal{S}')) \times c'_{b'}(b', x', n', \mathcal{S}') \\
& + u'(c'(b', x', n', \mathcal{S}'), 1) (p_{b'}^{s'}(b', x', n', \mathcal{S}')(1 - \mathbf{1}'\phi') \\
& + \mu'_{b'}(b', x', n', \mathcal{S}') \kappa' Q'(b', x', n', \mathcal{S}') + \mu'(b', x', n', \mathcal{S}') \kappa' Q'_{b'}(b', x', n', \mathcal{S}'))
\end{aligned} \tag{70}$$

$$\hat{\Omega}'_{b'} = R u''(c'(b', x', n', \mathcal{S}'), 1) \times c'_{b'}(b', x', n', \mathcal{S}') \tag{71}$$

Multiplying by R and using the FOC for consumption gives:

$$u'(c, 1) = \beta R \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left(u'(c', 1) - u''(c', 1) (\xi^{*'} p'^s + \omega^{*'}) + \xi^* \tilde{\Omega}'_{b'} \right) + u''(c, 1) \xi^* p^s + \mu^* \tag{72}$$

The first-order conditions for x' yields:

$$-\lambda^* p^x + \beta \mathbb{E}_{S'|S} \left[(\lambda'^* + \mu'^* \kappa') \mathbf{1} \phi' + \lambda'^* T_{x'}^{ins} + \Psi'^* MPI^{con'} \Lambda_{x'}^{n'} + \xi^* \tilde{\Omega}'_{x'} \right] - \Psi^* = 0 \quad (73)$$

where

$$\begin{aligned} \Lambda_{x'}^{n'} &= p^x (1 + R'^k) - \mathbf{1}' \phi' p^r - \zeta^1 \zeta^2 x' \zeta^{2-1} \\ \tilde{\Omega}'_{x'} &= u''(c'(b', x', n', S'), 1) (\nu_s g'(1) + Q'(b', x', n', S') + \mu'(b', x', n', S') \kappa' Q'(b', x', n', S')) \\ &\quad \times (c'_{x'}(b', x', n', S') + c'_{n'}(b', x', n', S') \Lambda_{x'}^{n'}) \\ &\quad + u'(c'(b', x', n', S'), 1) \left((p_{x'}^{s'}(b', x', n', S') + p_{n'}^{s'}(b', x', n', S') \Lambda_{x'}^{n'}) (1 - \mathbf{1}' \phi') \right. \\ &\quad \left. + (\mu'_{x'} + \mu'_{n'} \Lambda_{x'}^{n'}) (b', x', n', S') \kappa' Q'(b', x', n', S') \right. \\ &\quad \left. + \mu'(b', x', n', S') \kappa' (Q'_{x'}(b', x', n', S') + Q'_{n'}(b', x', n', S') \Lambda_{x'}^{n'}) \right) \end{aligned} \quad (74)$$

Rewriting gives:

$$p^x \lambda^* = \beta \mathbb{E}_{S'|S} \left[(\lambda'^* + \mu'^* \kappa') \mathbf{1} \phi' + \Psi'^* MPI^{con'} \Lambda_{x'}^{n'} + \xi^* \hat{\Omega}'_{x'} \right] - \Psi^* \quad (75)$$

The first-order condition for the house price p^s gives:

$$\mu^* \kappa (1 - \mathbf{1} \phi) - \xi^* u'(c, 1) = 0 \quad (76)$$

Solving for ξ^* gives:

$$\xi^* = \frac{\mu^* \kappa (1 - \mathbf{1} \phi)}{u'(c, 1)} \quad (77)$$

The first-order condition for the insurance price p^x gives:

$$-\lambda^* (x' + p^x \frac{\partial x'}{\partial p^x}) + MB(x') \frac{\partial x'}{\partial p^x} - \Psi^* \zeta p^r \frac{\partial x'}{\partial p^x} + \mathbb{E}_{S'|S} \left[\Psi'^* MPI^{con'} \Lambda_{p^x}^{n'} \right] = 0 \quad (78)$$

where $MB(x')$ is the marginal benefit of insurance when the capital constraint is not binding (i.e., the righthandside of equation (76) when $\Psi_t^* = 0$):

$$MB(x') = \beta \mathbb{E}_{S'|S} \left[(\lambda'^* + \mu'^* \kappa') \mathbf{1} \phi' + \Psi'^* MPI^{con'} \Lambda_{x'}^{n'} + \xi^* \hat{\Omega}'_{x'} \right] \quad (79)$$

Rewriting gives:

$$\underbrace{(MB(x') - \lambda^* p^x - \Psi^* \zeta p^r)}_{=0} \frac{\partial x'}{\partial p^x} = \lambda^* x' - \mathbb{E}_{S'|S} \left[\Psi'^* MPI^{con'} \Lambda_{p^x}^{n'} \right] \quad (80)$$

10.4 Optimal taxation

$$p_t^x u'(c, 1)(1 + \tau^x) = \beta \mathbb{E}_{S'|S} [(\mathbf{1}' \phi' p^r (u'(c', 1) + \mu' \kappa'))] \quad (82)$$

One has to find τ^x that yields the first order condition of the social planner.

This approach proceeds by inserting as p_t^x the valuation of insurance by the planner. First rewrite the insurance price as the ratio of expected future marginal benefits to marginal costs today:

$$p_t^x (1 + \tau^x) = \frac{\beta \mathbb{E}_{S'|S} [MB_{t+1}^{DE}]}{MC_t^{DE}} \quad (83)$$

Inserting for p_t^x the insurance valuation by the planner yields:

$$\frac{\beta \mathbb{E}_{S'|S} [MB_{t+1}^{PL}]}{MC_t^{PL}} (1 + \tau^x) = \frac{\beta \mathbb{E}_{S'|S} [MB_{t+1}^{DE}]}{MC_t^{DE}} \quad (84)$$

Solving for the tax yields:

$$\tau^x = \frac{\beta \mathbb{E}_{S'|S} [MB_{t+1}^{DE}]}{MC_t^{DE}} \frac{MC_t^{PL}}{\beta \mathbb{E}_{S'|S} [MB_{t+1}^{PL}]} - 1 \quad (85)$$

The insurance tax thus the ratio of the marginal benefit of insurance in the unconstrained's economy to the marginal benefit of insurance in the planner's economy corrected for differences in the marginal cost of insurance today. Equivalently, the tax equals the ratio of the insurance price in the DE to the insurance price in the planner's economy minus 1. Whenever the implementability and feasibility constraints today do not bind, one can derive the macro-prudential (MP) tax on insurance:

$$\tau_{MP}^x = \frac{\beta \mathbb{E}_{S'|S} [(u'(c', 1) + \mu' \kappa') \mathbf{1}' \phi' p^r]}{\beta \mathbb{E}_{S'|S} [(\lambda'^* + \mu'^* \kappa') \mathbf{1}' \phi' p^r + \Psi'^* MPI^{con'} \Lambda_{x'}^{n'}]} - 1 \quad (86)$$

Using similar steps, one can derive the optimal tax on borrowing as

$$\tau^b = \frac{\beta R \mathbb{E}_{S'|S} (u'(c', 1) - \xi^{*'} u''(c', 1) p'^s + \xi^* \hat{\Omega}_{b'}') + u''(c, 1) \xi^* p^s}{\beta R \mathbb{E}_{S'|S} u'(c', 1)} - 1 \quad (87)$$

Note that when the tax is implemented, the allocations of the planner and in the DE converge, so that $\mu = \mu^*$.

10.5 Computational Algorithm - decentralized equilibrium

The algorithm follows the approach of [Mendoza and Villalvazo \(2020\)](#) and solves the model's equilibrium conditions in recursive form. Denote $\mathcal{S} = \{\mathbf{1}, \theta, R\}$ as the triplet of exogenous shocks at date t , which includes the disaster indicator, disaster severity and the exogenous interest rate. The two endogenous state variables for the household are borrowing b and insurance x . The state variable for the insurer is net worth, which also enter the household problem. The recursive equilibrium is defined as a set of recursive functions for allocations $[c(b, x, n, \mathcal{S}), h(b, x, n, \mathcal{S}), b'(b, x, n, \mathcal{S}), x'(b, x, n, \mathcal{S})]$, prices $[p^x(b, x, n, \mathcal{S}), p^s(b, x, n, \mathcal{S})]$, and multiplier $[\mu(b, x, n, \mathcal{S})]$.

- **Step 0:** Start iteration j with an initial conjecture of the pricing functions $\hat{p}_j^x(b, x, n, \mathcal{S})$, $\hat{p}_j^s(b, x, n, \mathcal{S})$, decision rule for bonds $\hat{b}'_j(b, x, n, \mathcal{S}) = b$, and the multiplier $\hat{\mu}_j(b, x, n, \mathcal{S}) = 0$. Iterate on the insurer optimality condition (??) until the return on assets R^k , marginal values ν^a , ν^x and allocations ψ , Ψ converge. Compute insurance $x'_j(b, x, n, \mathcal{S})$ and asset quantities $a'_j(b, x, n, \mathcal{S})$ using the insurer leverage conditions. Compute equilibrium wages using insurer investments, and the household's and non-financial firms' labor optimality conditions.
- **Step 1:** Compute $l_j(b, x, n, \mathcal{S})$ using the first-order condition for the within-period loan and $\hat{\mu}_j$ and $\hat{c}_j$. Using $x'_j(b, x, n, \mathcal{S})$ and pricing functions, consumption then follows from the resource constraint:

$$c_j(b, x, n, \mathcal{S}) = b + w(b, x, n, \mathcal{S}) + \hat{p}_j^s(b, x, n, \mathcal{S})(1 - \mathbf{1}\phi) + \mathbf{1}\phi x + l_j(b, x, n, \mathcal{S}) - f(l_j(b, x, n, \mathcal{S})) - \frac{\hat{b}'_j(b, x, n, \mathcal{S})}{R} - \hat{p}_j^s(b, x, n, \mathcal{S}) - \hat{p}_j^x(b, x, n, \mathcal{S})x'(b, x, n, \mathcal{S}) \quad (88)$$

Note that housing drops out given the market clearing condition $s = 1$.

- **Step 2:** Assume the collateral constraint does not bind. Solve for $c_{j+1}(b, x, n, \mathcal{S})$ using the Euler equation:

$$c_{j+1}(b, x, n, \mathcal{S}) = \left\{ \beta R E_t \left[\left(c_j(\hat{b}'_j(b, x, n, \mathcal{S}), x'(b, x, n, \mathcal{S}), \mathcal{S}') + \nu_s g(1) \right)^{-\psi} \right] \right\}^{-\frac{1}{\psi}} - \nu_s g(1) \quad (89)$$

When computing $c_j(\cdot)$ and $h_j(\cdot)$, use interpolation if $\hat{b}'_j(b, x, n, \mathcal{S})$ and $x'(b, x, n, \mathcal{S})$ are intermediate points that are not on the grid, and extrapolation if $x'(b, x, n, \mathcal{S})$ or $\hat{b}'_j(b, x, n, \mathcal{S})$ exceed the boundaries of the grid.

- **Step 3:** Solve for $b'_{j+1}(b, x, n, \mathcal{S})$ using the resource constraint:

$$b'_{j+1}(b, x, n, \mathcal{S}) = R \left\{ b + w(b, x, n, \mathcal{S}) + \hat{p}_j^s(b, x, n, \mathcal{S}) (1 - \mathbf{1} \phi) + \mathbf{1} \phi x + l_j(b, x, n, \mathcal{S}) - f(l_j(b, x, n, \mathcal{S})) \right. \\ \left. - c_{j+1}(b, x, n, \mathcal{S}) - \hat{p}_j^s(b, x, n, \mathcal{S}) - \hat{p}_j^x(b, x, n, \mathcal{S}) x'(b, x, n, \mathcal{S}) \right\} \quad (90)$$

- **Step 4:** Evaluate if the collateral constraint binds. If

$$\kappa_t \left(\hat{p}_j^s(b, x, n, \mathcal{S}) (1 - \mathbf{1}_t \phi_t) + \mathbf{1}_t \phi_t x - l_j(b, x, n, \mathcal{S}) \right) + \frac{b'_{j+1}(b, x, n, \mathcal{S})}{R_t} \geq 0 \quad (91)$$

the constraint does not bind and go to Step 6. Otherwise, the collateral constraint is binding and go to Step 5.

- **Step 5:** Solve for $b'_{j+1}(b, x, n, \mathcal{S})$ by evaluating the collateral constraint with equality:

$$b'_{j+1}(b, x, n, \mathcal{S}) = -R_t \left\{ \kappa_t \left(\hat{p}_j^s(b, x, n, \mathcal{S}) (1 - \mathbf{1}_t \phi_t) + \mathbf{1}_t \phi_t x - l_j(b, x, n, \mathcal{S}) \right) \right\} \quad (92)$$

Determine $c_{j+1}(b, x, n, \mathcal{S})$ directly from the resource constraint:

$$c_{j+1}(b, x, n, \mathcal{S}) = b + w(b, x, n, \mathcal{S}) + \hat{p}_j^s(b, x, n, \mathcal{S}) (1 - \mathbf{1} \phi) + \mathbf{1} \phi x \\ - \frac{b'_{j+1}(b, x, n, \mathcal{S})}{R} - \hat{p}_j^s(b, x, n, \mathcal{S}) - \hat{p}_j^x(b, x, n, \mathcal{S}) x'(b, x, n, \mathcal{S}) + l_j(b, x, n, \mathcal{S}) - f(l_j(b, x, n, \mathcal{S})) \quad (93)$$

Step 6: Solve for $\mu_{j+1}(b, x, n, \mathcal{S})$ using the bonds Euler equation:

$$\mu_{j+1}(b, x, n, \mathcal{S}) = \left(c_{j+1}(b, x, n, \mathcal{S}) + \nu_s g(1) \right)^{-\psi} \\ - \beta R \mathbb{E} \left[\left(c_j(\hat{b}'_j(b, x, n, \mathcal{S}), x'(b, x, n, \mathcal{S}), \mathcal{S}') + \nu_s g(1) \right)^{-\psi} \right] \quad (94)$$

- **Step 7:** Update the insurance and house price:

$$p_{j+1}^x(b, x, n, \mathcal{S}) = \frac{\beta \mathbb{E}_t \left[\mathbf{1}_{t+1} \phi_{t+1} \left(\mathcal{U}_{j+1}(\hat{b}'_{j+1}, x', n', \mathcal{S}')^{-\psi} + \mu_{j+1}(\hat{b}'_{j+1}, x', \mathcal{S}') \kappa' \right) \right]}{\mathcal{U}_{j+1}(b, x, n, \mathcal{S})^{-\psi}} \quad (95)$$

$$\begin{aligned}
& p_{j+1}^s(b, x, n, \mathcal{S}) \\
&= \beta \mathbb{E}_t \left[\frac{\mathcal{U}_{j+1}(\hat{b}'_{j+1}, x', n', \mathcal{S}')^{-\psi}}{\mathcal{U}_{j+1}(b, x, n, \mathcal{S})^{-\psi}} \left\{ \nu_s g'(1) + \hat{p}_j^s(\hat{b}'_{j+1}(b, x, n, \mathcal{S}), x'(b, x, n, \mathcal{S}), \mathcal{S}')(1 - \mathbf{1}_{t+1}\phi_{t+1}) \right\} \right. \\
& \quad \left. + \frac{\mu_{j+1}(\hat{b}'_{j+1}, x', \mathcal{S}') \kappa'}{\mathcal{U}_{j+1}(b, x, n, \mathcal{S})^{-\psi}} \left\{ \hat{p}_j^s(\hat{b}'_{j+1}(b, x, n, \mathcal{S}), x'(b, x, n, \mathcal{S}), \mathcal{S}')(1 - \mathbf{1}_{t+1}\phi_{t+1}) \right\} \right] \quad (96)
\end{aligned}$$

where

$$\mathcal{U}_{j+1}(\hat{b}'_{j+1}, x', n', \mathcal{S}') = c_{j+1}(\hat{b}'_{j+1}(b, x, n, \mathcal{S}), x'(b, x, n, \mathcal{S}), \mathcal{S}') + \nu_s g(1) \quad (97)$$

$$\mathcal{U}_{j+1}(b, x, n, \mathcal{S}) = c_{j+1}(b, x, n, \mathcal{S}) + \nu_s g(1) \quad (98)$$

$$(99)$$

- **Step 8:** Compute the difference between the old and new insurance prices. Update the insurance price if the difference is not smaller than a lower than an upper bound, and return to step 0.

10.6 Computational Algorithm - planner's equilibrium

The algorithm to solve the planner's problem is similar to [Bianchi and Mendoza \(2018\)](#), but extends with insurance intermediaries.

- **Step 0:** Start iteration j with an initial conjecture of the pricing functions $\hat{p}_j^x(b, x, n, \mathcal{S})$, $\hat{p}_j^s(b, x, n, \mathcal{S})$, decision rule for bonds $\hat{b}'_j(b, x, n, \mathcal{S})$, insurance $\hat{x}'_j(b, x, n, \mathcal{S})$, consumption $\hat{c}_j(b, x, n, \mathcal{S})$, wages $\hat{w}_j(b, x, n, \mathcal{S})$, the multiplier $\hat{\mu}_j(b, x, n, \mathcal{S}) = 0$ and initial value function $\hat{V}_j(b, x, n, \mathcal{S})$. I use as initial functions the policies of the decentralized equilibrium.
- **Step 1:** Using the guess $\hat{p}_j^x$, compute insurance supply, wages and return on assets by iterating on the insurer problem.
- **Step 2:** Given $\hat{V}_j$, $\hat{x}'_j$, $\hat{p}_j^s$, $\hat{w}_j$, solve for the value and policy functions:

$$\mathcal{V}(b, x, n, \mathcal{S}) = \max_{c, b', s, x', \mu} u(c, s) + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \mathcal{V}(b', x', n', \mathcal{S}') \quad (100)$$

subject to

$$c + \frac{b'}{R} + p^x x' p^r = b + h w + \mathbf{1} \phi [x p^r - p^s] + l - f(l) \quad (101)$$

$$n' = a(1 + R'^k) - \mathbf{1}'\phi'x'p^r \quad (102)$$

$$\psi n = x' \quad (103)$$

$$\frac{b'}{R} \geq -\kappa [p^s (1 - \mathbf{1}\phi) + \mathbf{1}\phi x - l] \quad (104)$$

$$\mu \left(\frac{b'}{R} + \kappa [p^s (1 - \mathbf{1}\phi) + \mathbf{1}\phi x - l] \right) = 0 \quad (105)$$

$$\begin{aligned} p^s u'(c, 1) = \beta \mathbb{E} & \left[u'(c'(b', x', n', \mathcal{S}'), 1) \right. \\ & \times \left\{ \nu_s g'(1) + p^{s'}(b', x', n', \mathcal{S}') (1 - \mathbf{1}'\phi') \right\} \\ & \left. + p^{s'}(b', x', n', \mathcal{S}') (1 - \mathbf{1}'\phi') \mu'(b', x', n', \mathcal{S}') \kappa' \right] \end{aligned} \quad (106)$$

This problem is solved using value function iteration with linear interpolation when the insurance, bond and insurer net worth positions are outside the grid. First assume the collateral constraint is not binding. Use a Newton type algorithm to find the candidate solutions for $\hat{b}'_{j+1}$. Solve for $\hat{c}_{j+1}$ using the budget constraint. I solve for policies using a smooth optimizer. The key distinction with the decentralized equilibrium is that the planner endogenously updated insurer net worth in the next-period's value function.

- **Step 3:** Check if the collateral constraint (104) is binding. If binding, solve for each possible b'_{j+1} , the combinations μ , c , p^s and p^x that make hold the conditions hold with equality. Pick the combination that has the maximum value.
- **Step 4:** Repeat steps 1-3 until all functions convergence.